
Reliable Causal Estimation under Latent Markov Confounding

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Abstract

Causal inference from dependent observational time series is complicated when an unobserved, persistent state jointly affects treatment assignment and outcomes. Standard Double Machine Learning (DML) controls for observed covariates through orthogonal residualization, but does not by itself resolve confounding induced by an imperfect proxy for latent state dynamics. We study causal estimation under latent Markov confounding and characterize how latent-state uncertainty and downstream task conditioning propagate into causal error. We develop a regime-aware orthogonal estimator combining probabilistic state inference, temporally purged cross-fitting, and regularized inversion of the cross-regime score Jacobian. Our analysis separates nuisance-estimation error, latent-state recovery error, and variability induced by persistent regime occupation, and gives conditions under which the estimator approaches the corresponding oracle-state estimator as latent-state uncertainty decreases. We evaluate these properties using real-data-calibrated semi-synthetic time series constructed from high-frequency atmospheric observations across heterogeneous Indian cities, preserving persistence, seasonality, missingness, and regime overlap while keeping causal effects known. Across controlled variations in state persistence and observability, the experiments show how latent-state uncertainty predicts causal bias and when regime-aware estimation improves over standard and temporally blocked DML. Real sensor networks provide an observational stress test across four megacities spanning distinct

residual treatment variance scales and telemetry artifacts; falsification checks are reported explicitly where temporal causal interpretation is unsupported. Together, the results show that reliable causal estimation with latent sequential confounding requires evaluating representation uncertainty jointly with the information geometry of the downstream task.

1 INTRODUCTION

Many observational time series are driven by a persistent state that nobody measures: the boundary-layer regime of an airshed, the liquidity regime of a market, the operating mode of a plant. When that state moves both a treatment and an outcome, adjusting for observed covariates does not identify the effect. The standard remedy is a two-stage recipe: infer the state with a hidden Markov model (HMM) from auxiliary proxies, then estimate regime-aware effects with Double Machine Learning (DML; Chernozhukov et al., 2018). The recipe comes with an implicit reliability rule: *trust the estimate when the posterior is confident and the score Jacobian is well conditioned*. Both quantities are routinely reported, and both appear in error bounds of the form $C\varepsilon_\gamma/\lambda_{\min}(\mathbf{J})$, which we also prove (Theorem 3).

This paper shows that the rule fails in a specific and checkable way. A *generated* state does not only add noise to the downstream estimating equation; it changes that equation’s geometry. Mixing regimes leaks between-regime treatment contrasts into the residualized treatment, so the smallest eigenvalue of the generated-state Gram matrix can *grow* as the state estimate gets worse. In a controlled two-regime design it grows 70× while the causal error first rises and then falls (Figure 1a–b), and with posteriors learned by EM the generated-state matrix looks better conditioned than the oracle-state matrix in 82% of runs (Figure 1c). We call this *phantom causal resolution*: the estimate becomes less trustworthy while its standard conditioning diagnostic becomes more reassuring.

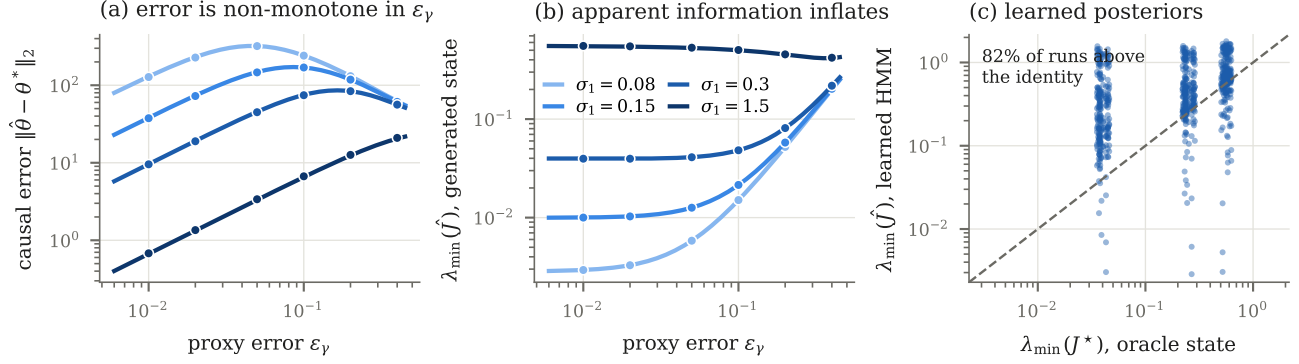


Figure 1: **Phantom causal resolution.** (a) Coupled regime-score error against proxy error ε_γ for four treatment-innovation scales σ_1 (dots: median of 100 runs of the 7×7 factorial design; lines: the exact population law of Proposition 1, with no fitted parameters). Error is non-monotone in ε_γ , peaking near $\varepsilon^* \approx 2\sigma/|\Delta_m|$. (b) The smallest eigenvalue of the generated-state Gram matrix in the same cells: unless σ_1 is large, it grows with proxy error ($70\times$ at $\sigma_1 = 0.08$), so a worse state estimate looks *better* conditioned. (c) With posteriors learned by EM from noisy proxies (720 runs), $\lambda_{\min}(\hat{J})$ exceeds the oracle-state value in 82% of runs (median ratio 1.9, 90th percentile 11).

The mechanism has a closed form, and the closed form says how to build the estimator. For the coupled regime score we derive the exact population displacement under state confusion (Proposition 1). It reproduces all 42 cells of a 4,900-run factorial experiment with observed/predicted ratio 0.999 (interquartile range 0.993–1.008), and it locates the damage in the nuisance step: *weighting* regime nuisances by the posterior leaks regime contrasts into the score. *Conditioning* the nuisances on the posterior instead makes the coupled score exactly Neyman-orthogonal for every posterior (Lemma 1). What remains is an exact omitted-variable identity in the *residual state variance* v that the posterior leaves behind (Theorem 4). For a pooled effect this is $\Delta_T \Delta_g v / (\Delta_T^2 v + \sigma^2)$; regime effects obey the same law in matrix form. The conditioning diagnostic shares a factor with the bias numerator, so no function of $(\varepsilon_\gamma, \lambda_{\min})$ alone can certify reliability. What is needed instead is the residual state information and the regime shifts it multiplies. Both are estimable up to one unidentified outcome shift, which supports a sensitivity analysis.

Contributions. (i) *Failure and limits:* an exact decomposition of the omitted-regime bias of standard DML, including bias amplification, and non-identification when proxies carry no state information (Theorems 1–2). (ii) *Mechanism:* exact laws for how state confusion enters coupled and posterior-conditioned estimators. They explain phantom resolution, the non-monotonicity of error, and why a posterior helps as an adjustment feature but hurts as a nuisance weight (Proposition 1, Theorem 4). (iii) *Estimator:* overlap-aware regime DML (OR-DML) with posterior-adjusted nuisances, purged block cross-fitting

and spectral regularization. It comes with exact orthogonality, a degradation bound separating nuisance from state-recovery error, and asymptotic normality including HMM-parameter and regime-occupation variance (Lemma 1, Theorems 3, 5). Across 900 synthetic datasets it is the most accurate regime-aware estimator, up to ties, at every proxy separation and persistence level we test. The posterior-weighted design of earlier drafts loses to hard regime fixed effects and is reported as an ablation. (iv) *Evidence:* a factorial design verified against the closed form; learned-posterior experiments that remove its circularity; a real-data-calibrated semi-synthetic benchmark on four Indian cities with real gaps and calibrated persistence and overlap; and an audited sensor stress test with aligned placebos and robustness values.

Related work. Orthogonal and debiased estimation (Chernozhukov et al., 2018, 2022b; Foster and Syrgkanis, 2023; Nie and Wager, 2021) and its time-series extensions (Lewis and Syrgkanis, 2020; Semenova et al., 2023; Ciganovic et al., 2026) assume the confounders are observed. Sequential deconfounders learn latent confounders with neural representations (Bica et al., 2020; Hatt and Feuerriegel, 2021); our results caution that representation quality and downstream reliability are different objects. Latent-state and proximal approaches target identification (Miao et al., 2018; Tchetgen Tchetgen et al., 2020; Mastouri et al., 2021; Cui et al., 2024; Clouth et al., 2025), and regime-switching models go back to Hamilton (1989); we instead quantify what a *plug-in* posterior leaves behind. Our bias law is an omitted-variable identity (Greenland, 1980; Cinelli and Hazlett, 2020; Chernozhukov et al., 2022), and its non-monotonicity is the algebra of bias amplification

(Pearl, 2010; Ding et al., 2017); we claim neither as new. What is new is its calibrated, dependent-data and regime-coupled form, the exact law for regime-weighted scores, the orthogonal design that follows from them, and the consequence that the Gram-matrix diagnostics used for weak overlap (Crump et al., 2009; D’Amour et al., 2021) can move in the wrong direction when the state is generated. Posterior-based adjustment is known to fail without identification (Wang and Blei, 2019; D’Amour, 2019); we characterize how it fails.

2 SETTING AND THE FAILURE OF STANDARD DML

We observe $(Y_t, T_t, X_t, Z_t)_{t=1}^N$ generated by

$$S_t \sim \text{Markov}(\Pi, \pi_0), \quad Z_t = \mu_Z(S_t) + \xi_t, \quad (1)$$

$$T_t = m(X_t, S_t) + V_t, \quad Y_t = \theta(S_t)T_t + g(X_t, S_t) + U_t, \quad (2)$$

with $S_t \in \{1, \dots, K\}$ latent, $\mathbb{E}[V_t | X_t, S_t] = 0$, $\mathbb{E}[U_t | X_t, S_t, T_t] = 0$ and $\theta(S_t) = \sum_k \theta_k^* \mathbb{I}(S_t = k)$. Targets are the regime effects θ_k^* and the stationary average $\theta_{\text{ATE}}^* = \sum_k \pi_k^* \theta_k^*$. In our application T is NO_2 , Y is $\text{PM}_{2.5}$, X collects lagged pollutants and weather, and Z is meteorology. Regime-conditional unconfoundedness, overlap and positivity, bounded variables, geometric α -mixing and the usual $\mathcal{O}_P(N^{-1/4})$ nuisance rates are stated as Assumptions 1–4 in Appendix A.

Theorem 1 (Omitted regime bias). *Standard DML that residualizes on X_t alone satisfies $\hat{\theta}_{\text{naive}} \rightarrow_p \sum_k \omega_k \theta_k^* + \mathcal{B}_{\text{het}} + \mathcal{B}_{\text{conf}}$, with $\omega_k = \mathbb{E}[\mathbb{I}(S_t = k) \tilde{T}_t^2] / \mathbb{E}[\tilde{T}_t^2]$ and $\mathcal{B}_{\text{conf}} = \sum_k \mathbb{E}[\pi_k(X) \Delta m_k(X) \Delta g_k(X)] / \mathbb{E}[\tilde{T}_t^2]$, where $\Delta m_k, \Delta g_k$ are regime deviations of the treatment mean and outcome baseline (full statement and proof in Appendix B).*

Because $\mathbb{E}[\tilde{T}_t^2]$ shrinks as X explains more of T while the latent covariance in the numerator does not, residualizing on more observed controls can *amplify* omitted-regime bias and flip its sign (Pearl, 2010). In our synthetic frontier, standard DML has bias 8.44 on a true ATE of 1.53 at every level of proxy quality.

Theorem 2 (Non-identification without state information). *If $P(Z_t | S_t = j) = P(Z_t | S_t = k)$ for all j, k and transitions are memoryless, there exist two models satisfying all assumptions, with identical laws of $(Y, T, X, Z)_{1:N}$ for every N and distinct, non-permuted effect vectors (Appendix C).*

Between these extremes lies the regime that matters in practice: proxies that are informative but imperfect. The rest of the paper is about that regime.

3 OVERLAP-AWARE REGIME DML

OR-DML infers posteriors $\gamma_{tk} = P(S_t = k | Z_{1:N})$ (smoothing) or $P(S_t = k | Z_{1:t})$ (causal filtering) with a Gaussian HMM. It cross-fits one pair of nuisances on the posterior-augmented features $F_t = (X_t, \gamma_t, \gamma_t \otimes X_t)$, $\hat{m}(F) \approx \mathbb{E}[T | F]$ and $\hat{\mu}(F) \approx \mathbb{E}[Y | F]$, on contiguous blocks separated by an embargo of τ^* steps. With $\tilde{T}_t = T_t - \hat{m}(F_t)$ and $\tilde{Y}_t = Y_t - \hat{\mu}(F_t)$ it solves the coupled system

$$\hat{J}_{jk} = \frac{1}{N} \sum_t \gamma_{tj} \gamma_{tk} \tilde{T}_t^2, \quad \hat{S}_k = \frac{1}{N} \sum_t \gamma_{tk} \tilde{T}_t \tilde{Y}_t, \quad (3)$$

$\hat{\theta}_\lambda = (\hat{J} + \lambda I)^{-1} \hat{S}$. The posterior enters twice: as an *adjustment feature* in the nuisances (fine resolution) and as a *weight* that defines the regime effects (coarse target). The embargo is the smallest lag after which all proxy autocorrelations fall below the Bartlett band, and at least $C \log N$, which makes the folds asymptotically independent under geometric mixing.

Lemma 1 (Exact orthogonality). *Let $\psi_k(\theta; m, \mu) = \gamma_k(T - m) \{Y - \mu - \sum_j \gamma_j \theta_j (T - m)\}$. At $m = \mathbb{E}[T | F]$, $\mu = \mathbb{E}[Y | F]$, the Gateaux derivative of $\mathbb{E}\psi_k$ in every direction (δ_m, δ_μ) of square-integrable functions of F vanishes, for every θ and every posterior γ ; so does that of $\mathbb{E}[\gamma_j \gamma_k (T - m)^2]$.*

Proof. Each derivative is an expectation of a product of an F -measurable factor (the weights γ and δ) and $T - m$ or $Y - \mu - \sum_j \gamma_j \theta_j (T - m)$, whose conditional mean given F is zero. $\square$

The natural alternative, used in earlier drafts of this work and closest to regime fixed effects, fits regime- k nuisances on X with weights $\gamma_{\cdot k}$. Its Gram part is not orthogonal for soft weights: under a nuisance perturbation its score derivative is 85% of the Jacobian, against 0.4% for Lemma 1 (Appendix E). Section 4 shows why it fails, and we keep it as an ablation. Two simpler estimators on the same inputs are our main baselines: *hard regime fixed effects* (one-hot arg max weights) and *posterior-feature DML*, a single-index DML with nuisances on F_t . Algorithm 1 summarizes the procedure and the diagnostics we recommend reporting with it. Complexity is $\mathcal{O}(B C_{\text{nuis}} + N K^2 + K^3)$; Appendix J checks $K \leq 5$.

4 HOW STATE UNCERTAINTY AND TASK CONDITIONING PROPAGATE

Let $\mathbf{H}_t = \mathbf{e}_{S_t}$, $\varepsilon_\gamma = \mathbb{E} \|\gamma_t - \mathbf{H}_t\|_1$, and let $\mathbf{J}, \mathbf{S}$ be the population versions of the coupled system at the population nuisances of either design. Define $\boldsymbol{\theta}_\gamma = \mathbf{J}^{-1} \mathbf{S}$ and $\mathbf{b}_\gamma = \mathbf{S} - \mathbf{J} \boldsymbol{\theta}^*$, so $\boldsymbol{\theta}_\gamma - \boldsymbol{\theta}^* = \mathbf{J}^{-1} \mathbf{b}_\gamma$ exactly.

Algorithm 1 OR-DML with a reliability report

Require: $\{Y_t, T_t, X_t, Z_t\}_{t=1}^N$, folds B , penalty $\lambda \geq 0$.

- 1: Fit a Gaussian HMM to Z (multi-restart EM); compute posteriors γ_t (smoothing or causal filter).
- 2: Embargo $\tau^* = \max\{h : \max_j |\hat{\rho}_j(h)| \geq 1.96/\sqrt{N}\} + 1$, at least $C \log N$.
- 3: Cross-fit $\hat{m}, \hat{\mu}$ on $F_t = (X_t, \gamma_t, \gamma_t \otimes X_t)$ over purged contiguous blocks.
- 4: Form $\hat{\mathbf{J}}, \hat{\mathbf{S}}$ by (3); return $\hat{\theta}_\lambda = (\hat{\mathbf{J}} + \lambda \mathbf{I})^{-1} \hat{\mathbf{S}}$ with a HAC sandwich.
- 5: **Report:** hard-FE and posterior-feature baselines; $\hat{v}, \hat{a}, \widehat{\text{Var}}(\tilde{T})$ and the robustness value b^* (Section 4); raw and standardized $\lambda_{\min}$; lead placebos aligned to the outcome's information set.

Theorem 3 (Graceful degradation). *Under Assumptions 2 and 4, suppose the nuisance targets are within $C_\eta \varepsilon_\gamma$ of the oracle regime nuisances in L^1 . This holds for posterior-weighted nuisances, and for posterior-adjusted nuisances when the posterior is calibrated given F and Z is independent of (T, Y) given (X, S) . If $\lambda_{\min}(\mathbf{J}) > 0$, then $\|\theta_\gamma - \theta^*\|_2 \leq C \varepsilon_\gamma / \lambda_{\min}(\mathbf{J})$ with explicit C , and $\|\hat{\theta}_\gamma - \theta^*\|_2 \leq C \varepsilon_\gamma / \lambda_{\min}(\mathbf{J}) + \mathcal{O}_P((N^{-1/2} + \rho_N) / \lambda_{\min}(\mathbf{J}))$, with $\rho_N = r_N^2$ for posterior-adjusted nuisances (Lemma 1) and $\rho_N = r_N$ for posterior-weighted ones, where r_N is the nuisance rate (Appendices D, E).*

The bound separates the two errors the abstract names: state recovery (ε_γ) and nuisance estimation (ρ_N). In a controlled design $\varepsilon_\gamma / \lambda_{\min}(\hat{\mathbf{J}})$ ranks causal error almost perfectly (Spearman 0.995 over 4,900 runs, versus 0.626 for ε_γ and 0.521 for $1/\lambda_{\min}$). The next two results explain why that success is fragile: $\lambda_{\min}(\mathbf{J})$ is itself a function of the state error.

The coupled score under state confusion. Consider two regimes with stationary weights π_s , treatment means m_s , outcome means $\mu_s = \theta_s m_s + g_s$ and innovation scales σ_s . Take regime-wise nuisances (the posterior-weighted design with correct regime means) and a proxy that places weight $1 - \alpha$ on the true regime and $\alpha = \varepsilon_\gamma / 2$ on the other.

Proposition 1 (Exact law for regime-weighted scores). *With $\gamma_j(s) = 1 - \alpha$ if $j = s$ and α otherwise,*

$$J_{jk} = \sum_s \pi_s \gamma_j(s) \gamma_k(s) [(m_s - m_j)(m_s - m_k) + \sigma_s^2],$$

$$S_j = \sum_s \pi_s \gamma_j(s) [(m_s - m_j)(\mu_s - \mu_j) + \theta_s \sigma_s^2].$$

In the symmetric case ($\pi_s = \frac{1}{2}$, $\sigma_s = \sigma$, $\Delta = m_1 - m_0$, $G = \mu_1 - \mu_0$), $\mathbf{J}$ has eigenvectors $(1, \pm 1)$ with eigenvalues $\lambda_+ = \frac{1}{2}\sigma^2 + \frac{1}{2}\alpha^2\Delta^2$ and $\lambda_- = \frac{1}{2}\sigma^2(1 -$

$2\alpha)^2 + \frac{1}{2}\alpha^2\Delta^2$, and

$$\theta_\gamma - \theta^* = \frac{\alpha \Delta G - \frac{1}{2}\alpha^2 \Delta^2 (\theta_0 + \theta_1)}{2\lambda_+} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \frac{(\theta_0 - \theta_1)(\alpha(1 - 2\alpha)\sigma^2 - \frac{1}{2}\alpha^2 \Delta^2)}{2\lambda_-} \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Corollary 1 (Phantom resolution and the error hump). *In the symmetric case: (a) $\lambda_{\min}(\mathbf{J}) = \lambda_-$ exceeds its oracle value $\sigma^2/2$ whenever $\varepsilon_\gamma > 8\sigma^2/(\Delta^2 + 4\sigma^2)$, so a confused state can look better conditioned than the true one; (b) when $|\Delta G| \gg \sigma^2|\theta_0 - \theta_1|$, the error is $\approx \sqrt{2} \frac{\varepsilon_\gamma}{2} |\Delta G|/(\sigma^2 + \varepsilon_\gamma^2 \Delta^2/4)$, maximized at $\varepsilon^* = 2\sigma/|\Delta|$ with peak $|G|/(\sqrt{2}\sigma)$, and decreasing in ε_γ beyond it; (c) the dominant displacement lies on the λ_+ eigenvector, so the worst-direction bound of Theorem 3 is loose by λ_+/λ_- .*

Proposition 1 is exact (verified symbolically in the supplement) and involves no sampling. Applied to the asymmetric factorial design it reproduces every cell's median error to within 0.98–1.02 $\times$ and locates the error peaks at $\varepsilon_\gamma \in \{0.05, 0.10, 0.20\}$ for $\sigma_1 \in \{0.08, 0.15, 0.30\}$, exactly where they are observed (Figure 1). The factorial's near-perfect rank correlation is therefore an algebraic identity of the design, not a statistical regularity. It works partly because $\lambda_{\min}(\hat{\mathbf{J}})$ inflates: replacing it by the design value σ_1^2 lowers the cell-level correlation from 0.997 to 0.958. The leading displacement $\alpha \Delta G$ is the product of state error with the regime contrasts that the weighted nuisances fail to remove.

Posterior-adjusted designs. Posterior-adjusted nuisances remove those contrasts up to what the posterior cannot explain. Let $v_t = \gamma_t(1 - \gamma_t)$ be the posterior variance of the regime indicator, and let $T_t = m(X_t) + aS_t + V_t$ and $Y_t = \theta_s T_t + g(X_t) + bS_t + U_t$.

Theorem 4 (Residual-state bias law). *Let $K = 2$ and suppose the posterior is calibrated given F ($P(S_t = 1 | F_t) = \gamma_t$), V_t is independent of (S, Z, X) with variance σ^2 , nuisances are cross-fitted at rates with $r_N^2 = o(N^{-1/2})$, and the posterior forgets distant proxies (Appendix E). With score weights $\mathbf{w}_t = (1 - \gamma_t, \gamma_t)$ (OR-DML) or $\mathbf{w}_t \equiv 1$ (posterior-feature DML),*

$$\hat{\theta} \rightarrow_p \bar{\theta} + \mathbf{J}^{-1} \mathbf{b}_\gamma,$$

$$\mathbf{J} = \mathbb{E}[\mathbf{w} \mathbf{w}^\top (a^2 v + \sigma^2)], \quad \mathbf{b}_\gamma = a \mathbb{E}[\mathbf{w} c v],$$

with $c_t = b + (\theta_1 - \theta_0)\{m(X_t) + a(1 - \gamma_t)\}$, $\bar{\theta} = \theta^$ for OR-DML and the $(a^2 v + \sigma^2)$ -weighted mean of $\theta_0 + (\theta_1 - \theta_0)\gamma$ for the pooled estimator. Hence: (a) with no treatment regime shift ($a = 0$) both are consistent for any calibrated posterior, whatever b and the effect heterogeneity; (b) for homogeneous effects the pooled bias is $abv/(a^2 v + \sigma^2)$, hump-shaped in a with peak at*

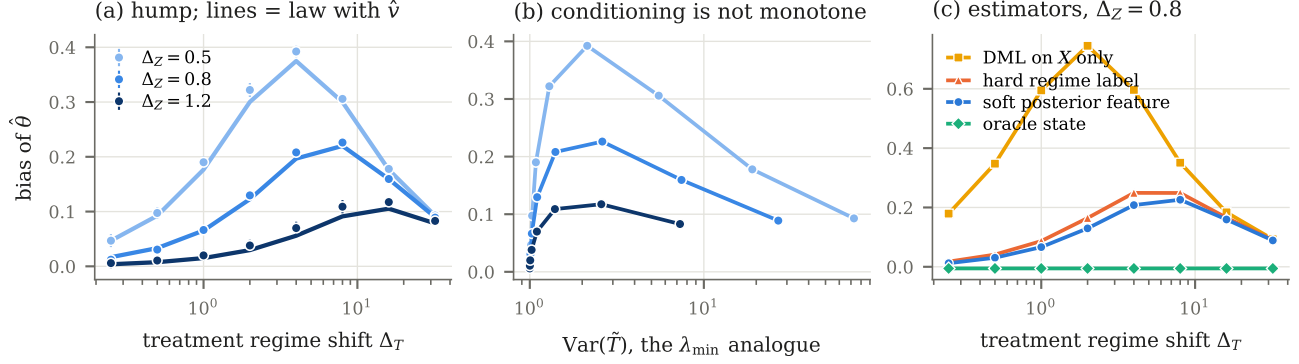


Figure 2: **The residual-state bias law with learned HMM posteriors** (pooled posterior-feature DML; smoother, linear nuisances, 12 replications per point, 95% bands). (a) Measured bias is hump-shaped in the treatment regime shift and matches the law evaluated with the observable plug-in $\hat{v}$ (lines; across all 72 cells, correlation 0.996, median relative error 5.2%). (b) The same bias against $\text{Var}(\tilde{T})$, the scalar analogue of $\lambda_{\min}(\hat{J})$: conditioning improves monotonically while bias first rises and then falls. (c) Soft posterior features beat hard regime labels, as information monotonicity predicts; DML on X alone is far worse; the oracle state is unbiased.

$|a| = \sigma/\sqrt{v}$ and $\sup |B| = |b|\sqrt{v}/(2\sigma)$, increasing in v , and a soft posterior feature is never worse than its hard label; (c) J grows with a^2v , so phantom resolution is not specific to the weighted design.

The law is exact up to sampling error: over 9 configurations with $N = 10^6$ the estimator and the law agree to within 0.008, and with $a = 0$ the bias is 0.005 (Appendix E). The pooled form with $K > 2$ regimes is $a^\top \Sigma b / (a^\top \Sigma a + \sigma^2)$ with Σ the residual state covariance (Appendix G). Both laws have the same anatomy. A numerator proportional to state error times the product of the treatment and outcome regime shifts is divided by a residual treatment variance that the same treatment shift inflates. This is why $\text{Var}(\tilde{T})$, the scalar analogue of $\lambda_{\min}(\hat{J})$, cannot be read as a monotone certificate (Figure 2). The two designs differ in what multiplies the state error: weighted nuisances leak the raw contrast ΔG at first order in ε_γ , adjusted nuisances only the residual variance v .

Observability and inference. For a calibrated posterior, $\hat{v} = N^{-1} \sum_t \hat{\gamma}_t (1 - \hat{\gamma}_t)$ satisfies $|\mathbb{E} \hat{v} - v| \leq \sqrt{CE} + CE$ in the L^2 calibration error CE , and the treatment shift a and $\text{Var}(\tilde{T})$ are estimable. Only the outcome shift b is unidentified, so the robustness value $b^* = |\hat{\theta}| \widehat{\text{Var}}(\tilde{T}) / \|\hat{\Sigma} \hat{a}\|$ (the smallest outcome shift that explains the estimate away) gives a one-parameter sensitivity analysis. For EM-fitted HMMs the plug-in converges at rate $N^{-1/2} + \sqrt{\delta_X}$, where δ_X is regime information carried by the controls (Proposition 3). Under misspecified emissions $\hat{v}$ is a lower bound.

Theorem 5 (Inference around the proxy target). *Under the assumptions of Theorem 3, geometric mixing, an embargo $\tau^* \geq C \log N$ with $C > 1/c_\alpha$, and HMM parameters estimated at rate $N^{-1/2}$ on a cali-*

bration window, $\sqrt{N}(\hat{\theta}_\lambda - \theta_\lambda) \rightarrow \mathcal{N}(0, \Sigma_\lambda^{\text{tot}})$ with $\Sigma_\lambda^{\text{tot}}$ including the HMM-parameter term, where $\theta_\lambda - \theta^ = (J + \lambda I)^{-1}(\mathbf{b}_\gamma - \lambda \theta^*)$ and $\|\mathbf{b}_\gamma\| \leq C\varepsilon_\gamma$. Intervals cover θ^* only if $\sqrt{N}(C\varepsilon_\gamma + \lambda M_\theta)/(\lambda_{\min} + \lambda) \rightarrow 0$, i.e. the estimator approaches the oracle-state estimator as $\varepsilon_\gamma = o(N^{-1/2})$. The occupation-weighted ATE has an extra variance factor $(1 + \rho)/(1 - \rho)$ from regime persistence ρ (Appendix H).*

When ε_γ is not negligible, nominal Wald tests are invalid. In a regime-level null test they reject in 78.0% of runs for the weighted design and 31.0% for hard fixed effects (Appendix L). This is why we pair every estimate with the sensitivity analysis rather than a point correction.

5 EXPERIMENTS

Learned posteriors break the scalar ratio. To remove the synthetic-proxy circularity, we learn posteriors with a Gaussian HMM (EM, three restarts, smoothing) from noisy proxies across $\Delta_Z \times \sigma_1 \times \Delta_T$ (36 cells, 720 runs, $N = 1,200$). All diagnostics use the realized generated-state Gram matrix. Table 1 shows that the ratio $\varepsilon_\gamma / \lambda_{\min}(\hat{J})$ falls from 0.995 to 0.517. Restoring the oracle geometry recovers 0.750. Restoring instead the numerator scale the laws predict, using the plug-in regime shifts $\hat{\Delta}_T, \hat{\Delta}_Y$ of treatment and outcome, gives 0.695 (0.863 across cells), and 0.664 with entropy in place of ε_γ , which is fully observable. With the same posteriors, hard fixed effects beat the posterior-weighted design in 100% of cells, while posterior-adjusted OR-DML beats hard fixed effects in 72% (median error ratio 0.97; 0.69 against the weighted design). All of its losses occur at the smallest treatment innovation, where the a^2v term dominates J .

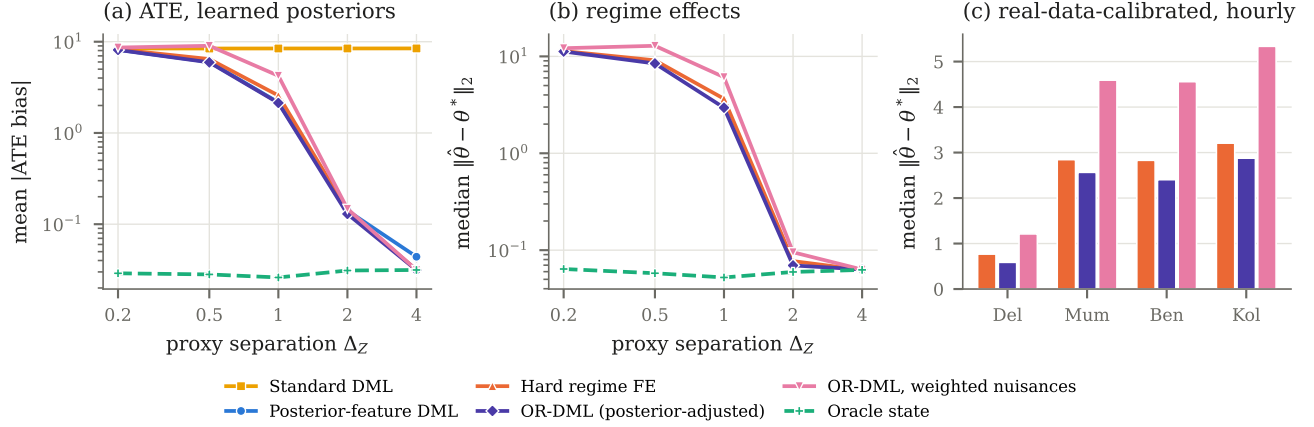


Figure 3: **Conditioning nuisances on the posterior, rather than weighting them by it, fixes the coupled estimator.** (a) Mean absolute ATE bias and (b) median regime-effect error along the observability frontier with learned HMM posteriors (100 datasets per point, persistence 0.88). Posterior-adjusted OR-DML is the most accurate regime-aware estimator at every Δ_Z (ties at $\Delta_Z = 4$); the posterior-weighted design is the least accurate wherever states are imperfectly recovered, as Proposition 1 predicts. Below $\Delta_Z \approx 1$ every proxy-based estimator fails. (c) Regime-effect error in the real-data-calibrated semi-synthetic benchmark (30 datasets per city): the same ordering in every city.

Table 1: Spearman correlation between failure scores and causal error. Synthetic proxy: factorial design. Learned: EM posteriors; “within Δ_T ” averages correlations at fixed treatment shift.

Failure score	synthetic proxy		learned HMM posteriors	
	runs		runs	within Δ_T cells
ε_γ (posterior error)	0.626		0.187	0.083 0.222
$1/\lambda_{\min}(\hat{J})$	0.521		0.465	0.483 0.509
$\varepsilon_\gamma/\lambda_{\min}(\hat{J})$	0.995		0.517	0.388 0.564
$\bar{H}/\lambda_{\min}(\hat{J})$ (observable)	—		0.581	0.442 0.611
$\varepsilon_\gamma/\lambda_{\min}(\hat{J}^*)$ (oracle geometry)	—		0.750	0.727 0.783
$\bar{H}/ \Delta_T \Delta_Y /\lambda_{\min}(\hat{J})$ (observable)	—		0.664	0.730 0.675
$\varepsilon_\gamma/ \Delta_T \Delta_Y /\lambda_{\min}(\hat{J})$	—		0.695	0.720 0.863

Observability and persistence. Table 2 (left) varies proxy separation Δ_Z in the DGP of Theorem 1; a persistence sweep adds regime-0 stay probabilities 0.70 and 0.97 (900 datasets in all). Standard and blocked DML are stuck at the omitted-regime bias. All proxy-based estimators fail at $\Delta_Z \leq 1$ and approach the oracle at $\Delta_Z \geq 2$. Among the 7 cells with $\Delta_Z \geq 1$ (Figure 3), OR-DML is best or tied for the ATE in 7 and for regime effects in 7. At $\Delta_Z = 2$ its ATE bias is 0.129 (hard fixed effects 0.141, weighted design 0.147) with coverage 91%, and at $\Delta_Z = 1$ its regime-effect error is 2.954 against 3.652 and 6.107. Persistence helps exactly through the posterior: at $\Delta_Z = 1$, raising the stay probability from 0.70 to 0.97 lowers ε_γ from 0.090 to 0.026 and OR-DML’s bias from 2.725 to 0.960 (hard fixed effects: 3.260 to 1.194). On the legacy version of this DGP the excess bias of soft conditioning over the oracle is predicted by the law to within 0.04 at every Δ_Z (Appendix G.1), so the transition from failure to success is explained by $v(\Delta_Z)$, not by the spectrum of

$\hat{J}$. Under a continuous latent mixture that no two-state model describes, OR-DML beats hard fixed effects in 100% of 100 replications, while the weighted design beats them in 0% (Appendix L).

Real-data-calibrated benchmark. The right half of Table 2 keeps everything real except the causal effects. That includes meteorological proxies and controls, the observed NO₂ series, the hourly missingness pattern, and regime persistence and overlap: true paths are drawn by forward-filtering backward-sampling from an HMM calibrated per city (stay probabilities 0.90–0.96). With posteriors that are exactly calibrated and 95–96% accurate, every proxy-based estimator retains large bias and zero coverage. For regime effects OR-DML is the most accurate regime-aware estimator in every city (Delhi 0.59, against 0.77 for hard fixed effects and 1.21 for the weighted design). For the pooled ATE it is best in Delhi and Kolkata, and posterior-feature DML is best in Mumbai and Bengaluru. The weighted design is worse than standard DML outside Delhi. The plug-in law, computed from the posterior, the fitted treatment shift and the injected outcome shift, ranks the four cities’ biases correctly, with magnitudes within 7–37%. The generated-state $\lambda_{\min}$ exceeds its oracle value by 1.7–2.4 $\times$ in Mumbai, Bengaluru and Kolkata (1.04 $\times$ in Delhi).

Representation quality is not task reliability. Across 15 shifted worlds and four encoders (HMM, GRU, Transformer, linear SSM), adding posterior NLL to state-recovery F_1 raises leave-one-world-out prediction of downstream reliability from $R^2 = 0.113$ to 0.221. Yet temperature scaling cuts GRU/Transformer ECE

Table 2: **Posterior-adjusted OR-DML is the most accurate regime-aware estimator, but no estimator removes residual-state bias.** *Left:* synthetic frontier with learned HMM posteriors (100 datasets per Δ_Z , $N = 1,200$, persistence 0.88, true ATE 1.53). *Right:* real-data-calibrated semi-synthetic benchmark on the corrected hourly grid (30 datasets per city, $N = 2,324-9,034$; real meteorology, NO_2 and gaps; true states drawn from the calibrated HMM’s smoothing distribution). Bold: best non-oracle within each block. Coverage, persistence variants and all estimators: Appendices L and M.

Estimator	synthetic, proxy separation Δ_Z			real-data-calibrated, hourly			
	1	2	4	Del	Mum	Ben	Kol
<i>ATE: mean absolute bias</i>							
Oracle-state	0.03	0.03	0.03	0.03	0.07	0.12	0.14
Standard DML	8.43	8.44	8.44	1.94	3.00	2.97	3.23
Block DML	8.47	8.47	8.48	2.16	3.09	3.20	3.30
DML + Z controls	5.43	2.63	0.86	1.22	3.03	2.60	3.06
Posterior-feature DML	2.16	0.14	0.04	0.55	1.03	1.51	2.17
Hard regime FE	2.57	0.14	0.03	0.55	1.86	1.97	2.30
OR-DML, weighted nuisances	4.24	0.15	0.03	0.88	3.10	3.22	3.81
OR-DML (posterior-adjusted)	2.13	0.13	0.03	0.42	1.65	1.69	2.05
<i>Regime effects: median $\ \hat{\theta} - \theta^*\ _2$</i>							
Hard regime FE	3.65	0.08	0.06	0.77	2.84	2.83	3.21
OR-DML, weighted nuisances	6.11	0.10	0.06	1.21	4.59	4.56	5.33
OR-DML (posterior-adjusted)	2.95	0.07	0.06	0.59	2.56	2.40	2.88
posterior error ε_γ	0.062	<0.001	<0.001	0.087	0.097	0.088	0.083

by 47.9%/60.3% without improving failure discrimination ($\Delta\text{AUC} -0.0035/-0.0075$). By Theorem 4, recalibration changes the *estimate* $\hat{v}$ but not the residual information the downstream task actually has (Appendix N).

6 OBSERVATIONAL STRESS TEST: FOUR CITIES

We apply OR-DML and the sensitivity analysis to $\text{NO}_2 \rightarrow \text{PM}_{2.5}$ in Delhi, Mumbai, Bengaluru and Kolkata (OpenAQ/CPCB stations and Open-Meteo weather, Feb 2025–Jun 2026) on a *corrected* hourly grid. An audit found that the legacy panel rounded half-hour timestamps with ties-to-even, collapsing it onto a two-hour grid, so “hourly” lags were two hours. This changed Delhi’s pooled estimate from 0.58 to 0.37 (Appendix O). Placebo leads are aligned to the outcome’s own information set. Without this alignment the one-hour lead is mechanically zero, because the lagged outcome is a control. Real posteriors are sharp here, so the weighted design gives the same ATE to within 0.12 outside Kolkata (Delhi 0.241).

Table 3 separates four axes the reliability rule conflates. *Scale:* raw $\lambda_{\min}$ ranges from 0.0012 to 63.2, but the standardized value only from 0.20 to 0.46. Raw conditioning reflects units and sensor resolution, not collinearity. *Data quality:* Kolkata’s NO_2 takes 16 distinct values, and Mumbai has a 277-hour constant run whose removal moves the ATE from 3.02 to 0.38 (N 8,113→6,256). *Temporal validity:* Bengaluru and

screened Kolkata fail lead placebos, the latter strongly ($p = 0.0003$), so the apparently significant screened Kolkata effect (3.32) is not interpretable. *Residual confounding:* Delhi’s association is positive in both regimes (0.430 and 0.102; ATE 0.234) and would need an outcome shift of 82 SD (90% interval 54–144) to be explained away through the HMM-regime channel. Its one-hour lead placebo nonetheless rejects (0.072, $p = 0.020$, about a third of the contemporaneous coefficient). This points to short-horizon confounding outside that channel. We therefore report Delhi as an exposure association that is robust to residual *regime* confounding, not as an emissions effect, and the other three cities as inconclusive.

7 DISCUSSION

The intuition we set out to test was that a confident latent state and a well-conditioned score Jacobian together certify a causal estimate. They do not. A generated state enters the downstream estimating equation twice, in its bias and in its apparent information, and the two move together. The scalar $\varepsilon_\gamma/\lambda_{\min}$ is an honest worst-case bound and an excellent ranking device when the state error is synthetic and the regime shifts are fixed. It degrades sharply once posteriors are learned, because its denominator is partly made of the error it is meant to measure. Two rules survive. *Design:* condition the nuisances on the posterior and use it only to weight the target. This makes the coupled score exactly orthogonal and gave the most accurate regime

estimates in every benchmark here. *Reporting:* state the residual state variance, the regime shifts it multiplies and a robustness value; compare against hard fixed effects and posterior-feature baselines; and test temporal validity with aligned placebos.

Outlook. Nothing in the mechanism is specific to HMMs. Any generated state (a neural encoder’s regime embedding, a clustering label, a changepoint segmentation) enters a downstream estimating equation both as an adjustment and as a source of apparent information, and the same accounting applies: what matters is the residual state variance the representation leaves, multiplied by the regime shifts of treatment and outcome, not the representation’s accuracy or the conditioning of the score it feeds. Evaluating representations for causal use therefore means reporting that residual, and designing estimators so that the representation is used as an orthogonal adjustment rather than as a weight on the nuisance fit.

Limitations. The laws are probability limits under linear regime shifts, exogenous treatment innovations and calibrated posteriors. The robustness value bounds rather than corrects, and is optimistic under misspecified emissions. Posterior-adjusted OR-DML does not remove residual-state bias: under weak observability every proxy-based estimator, ours included, has large bias and no coverage, and for the pooled ATE posterior-feature DML is sometimes more accurate. The city analysis is observational. Proxies overlap controls, lagged pollutants may be mediators, and Delhi’s lead placebo rejects.

Table 3: Four-city stress test on the corrected hourly grid (OR-DML, HAC lag 24 h). $\lambda_{\min}$ is raw, $\lambda_{\min}^{\text{std}}$ is scale-free. Lead placebo: smallest p -value over leads $h \in \{-6, -3, -1\}$ h. b^* : robustness value in SD of $\tilde{Y}$ (posterior-feature analysis; for screened rows, its own screened sample). Screened: runs of ≥ 6 identical NO_2 readings removed (no change for Delhi).

City	N	$\lambda_{\min}$	$\lambda_{\min}^{\text{std}}$	$\hat{\theta}_{\text{ATE}}$ (SE)	$\hat{\theta}_1$ (SE)	$\hat{\theta}_2$ (SE)	lead placebo	b^* (SD)
Delhi	9,024	63.2	0.46	0.234 (0.032)	0.430 (0.053)	0.102 (0.033)	rejects ($h = -1$, $p=0.020$)	82.3
Mumbai	8,113	0.0851	0.41	3.017 (1.682)	5.480 (2.417)	1.189 (2.429)	passes ($p_{\min}=0.33$)	11.9
Bengaluru	7,775	2.4	0.39	0.004 (0.074)	0.131 (0.140)	-0.132 (0.057)	rejects ($h = -6$, $p=0.016$)	26.8
Kolkata	2,322	0.00116	0.40	1.217 (1.141)	-0.674 (0.455)	3.894 (2.809)	passes ($p_{\min}=0.11$)	10.9
Mumbai (screened)	6,256	0.0495	0.20	0.383 (1.978)	-4.012 (4.688)	2.757 (1.496)	passes ($p_{\min}=0.05$)	1.2
Bengaluru (screened)	7,180	2.3	0.40	0.020 (0.082)	0.185 (0.155)	-0.156 (0.069)	rejects ($h = -3$, $p=0.014$)	35.3
Kolkata (screened)	1,422	0.00172	0.40	3.319 (1.397)	0.191 (0.532)	6.916 (2.983)	rejects ($h = -1$, $p=0.000$)	16.5

AI Use Statement

Large language models assisted with code, derivations, experiment scripting and drafting. Every number in this paper is generated from committed artifacts by a single script (`src/synthesis/make_tables_numbers.py`) and checked by `verify_submission.py`, which re-derives all tables and macros and fails on any mismatch. The closed forms are verified symbolically, and earlier drafts' errors found in audit (time grid, placebo alignment, overstated estimator claims) are documented in Appendix Q. The authors reviewed all arguments and take full responsibility for the content.

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CHECKLIST

1. For all models and algorithms presented, check if you include:
 - (a) A clear description of the mathematical setting, assumptions, algorithm, and/or model. **[Yes]** Sections 2–4, Appendix A, Algorithm 1.
 - (b) An analysis of the properties and complexity (time, space, sample size) of any algorithm. **[Yes]** Lemma 1, Theorems 3, 4, 5, 6; complexity and scalability in Appendix J.
 - (c) (Optional) Anonymized source code, with specification of all dependencies, including external libraries. **[Yes]** Supplementary archive with `requirements.txt`, `reproduce.sh` and `verify_submission.py`.
2. For any theoretical claim, check if you include:
 - (a) Statements of the full set of assumptions of all theoretical results. **[Yes]** Appendix A; Assumptions 5–7, 8, 9; setting (E) for learned HMMs.
 - (b) Complete proofs of all theoretical results. **[Yes]** Appendices B–I. The change-point scaling of the residual variance is summarized here and proved in the supplementary archive; it is not used for any main-text claim.
 - (c) Clear explanations of any assumptions. **[Yes]** Including where they fail on the real data ($\delta_X > 0$; multimodal likelihood for Mumbai; exclusion restrictions for proximal identification).
3. For all figures and tables that present empirical results, check if you include:
 - (a) The code, data, and instructions needed to reproduce the main experimental results. **[Yes]** Appendix P; raw sensor files are included.
 - (b) All the training details (e.g., data splits, hyperparameters, how they were selected). **[Yes]** Appendices K–O.
 - (c) A clear definition of the specific measure or statistics and error bars. **[Yes]** Mean absolute bias, RMSE and coverage over stated replications; HAC standard errors with stated lags; block-bootstrap intervals for robustness values.
 - (d) A description of the computing infrastructure used. **[Yes]** All experiments run on a two-core CPU machine; runtimes in Appendix P.
4. If you are using existing assets (e.g., code, data, models) or curating/releasing new assets:
 - (a) Citations of the creator if your work uses existing assets. **[Yes]** OpenAQ/CPCB and Open-Meteo.
 - (b) The license information of the assets, if applicable. **[Yes]** Public air-quality and reanalysis data under their providers’ open-data terms.
 - (c) New assets either in the supplemental material or as a URL, if applicable. **[Yes]** Code, processed data and all result files in the supplementary archive.
 - (d) Information about consent from data providers/curators. **[Not Applicable]** Aggregate environmental measurements.
 - (e) Discussion of sensible content if applicable, e.g., personally identifiable information or offensive content. **[Not Applicable]**
5. If you used crowdsourcing or conducted research with human subjects:
 - (a) The full text of instructions given to participants and screenshots. **[Not Applicable]**
 - (b) Descriptions of potential participant risks, with links to IRB approvals if applicable. **[Not Applicable]**
 - (c) The estimated hourly wage paid to participants and the total amount spent on participant compensation. **[Not Applicable]**

SUPPLEMENTARY MATERIAL

Roadmap. Appendix A states the assumptions. Appendices B–D prove Theorems 1–3. Appendix E treats posterior-adjusted nuisances: Lemma 1, Theorem 3 for that design, the proof of Theorem 4 and its numerical checks. Appendix F proves Proposition 1 and Corollary 1, and records the phantom-resolution results for hard classifiers and higher dimensions. Appendix G gives the pooled law for $K \geq 2$ regimes and its learned-HMM extension. Appendix H covers inference. Appendix I gives the operational surrogate, proxy screening and spectral regularization. Appendix J gives the algorithm and scalability. Appendices K–N document every experiment. Appendix O documents the sensor data and its audit. Appendix P explains how to reproduce and verify every number. Appendix Q records what changed relative to earlier drafts and why.

A ASSUMPTIONS

Assumption 1 (Regime-Conditional Unconfoundedness & Innovation Exogeneity). Conditional on observed covariates X_t , latent regime S_t , and proxy history $Z_{1:N}$, potential outcomes are unconfounded ($Y_t(u) \perp T_t \mid (X_t, S_t, Z_{1:N}), \forall u$) and structural innovations have zero mean:

$$\mathbb{E}[U_t \mid X_t, S_t, T_t, Z_{1:N}] = 0. \quad (4)$$

Assumption 2 (Overlap, Envelopes & State Positivity). There exist constants $\eta, c_0 > 0$, $B_T, B_Y < \infty$, and $c_1, c_2 > 0$ such that:

1. **State Positivity & Envelopes:** Stationary distribution satisfies $\min_k \pi_k^* \geq \eta$, and for almost every x and all k , $P(S_t = k \mid X_t = x) \geq c_0$, $\mathbb{E}[\gamma_{tk} \mid X_t = x] \geq c_0$. Variables satisfy $|T_t| \leq B_T$, $|Y_t| \leq B_Y$ a.s., so $|\tilde{T}_{tk}| \leq M_T \equiv 2B_T$, $|Y_{tk}| \leq M_Y \equiv 2B_Y$.
2. **Within-Regime Treatment Overlap:** For all $k \in \{1, \dots, K\}$ and almost everywhere on $\text{supp}(X_t)$:

$$c_1 \leq \text{Var}(T_t \mid X_t, S_t = k) \leq c_2. \quad (5)$$

Assumption 3 (Latent Proxy Sufficiency). The auxiliary state proxy sequence $Z_{1:N}$ satisfies latent proxy sufficiency: $P(S_t = k \mid X_t, T_t, Z_{1:N}) = P(S_t = k \mid Z_{1:N}) \equiv \gamma_{tk}$ (or $P(S_t = k \mid Z_{1:t})$ for causal forward filtering) almost surely for all $k \in \{1, \dots, K\}$. *Discussion.* That is, $Z_{1:N}$ screens off S_t from (X_t, T_t) . Combined with Assumption 1, this guarantees $\mathbb{E}[Y_t \mid X_t, T_t, Z_{1:N}] = \sum_{k=1}^K \gamma_{tk} [\theta_k^* T_t + g_k(X_t)]$. Notably, our simulation DGPs deliberately violate Assumption 3 ($S_t \not\perp T_t \mid Z_t$) via regime-dependent treatment means; crucially, Theorem 3 does not require Assumption 3, depending only on envelopes and rates (Assumptions 2 and 4).

Assumption 4 (Nuisance Estimation and Mixing Rates). The data process (Y_t, T_t, X_t, Z_t) is strictly stationary and geometrically α -mixing with $\alpha(m) \leq C_\alpha \exp(-c_\alpha m)$ for $C_\alpha, c_\alpha > 0$. Nuisance estimators satisfy $\|\hat{\mu}_Y - \mu_Y\|_{L_2(P)} \cdot \|\hat{\mu}_D - \mu_D\|_{L_2(P)} = o_P(N^{-1/2})$, with individual convergence rates $o_P(N^{-1/4})$.

B PROOF OF THEOREM 1

Full statement. Under (1)–(2), with $\tilde{T}_t = T_t - \mathbb{E}[T_t \mid X_t]$,

$$\mathcal{B}_{\text{het}} = \frac{\sum_k \theta_k^* \mathbb{E}[\bar{m}(X_t) \Delta m_k(X_t) \pi_k(X_t)]}{\mathbb{E}[\tilde{T}_t^2]}, \quad (6)$$

$$\mathcal{B}_{\text{conf}} = \frac{\sum_k \mathbb{E}[\pi_k(X_t) \Delta m_k(X_t) \Delta g_k(X_t)]}{\mathbb{E}[\tilde{T}_t^2]}, \quad (7)$$

with $\pi_k(X) = P(S_t = k \mid X_t)$, $\bar{m}(X) = \mathbb{E}[T_t \mid X_t]$, $\Delta m_k(X) = \mathbb{E}[T_t \mid X_t, S_t = k] - \bar{m}(X)$ and $\Delta g_k(X) = g(X, k) - \mathbb{E}[g(X_t, S_t) \mid X_t]$. Consider the Structural Causal Model (1)–(2):

$$\begin{aligned} T_t &= m(X_t, S_t) + V_t, \quad \mathbb{E}[V_t \mid X_t, S_t] = 0, \\ Y_t &= \sum_{k=1}^K \theta_k^* \mathbb{I}(S_t = k) T_t + g(X_t, S_t) + U_t, \quad \mathbb{E}[U_t \mid X_t, S_t, T_t] = 0. \end{aligned}$$

Define the unconditioned expectations: $\bar{m}(X_t) \equiv \mathbb{E}[T_t | X_t] = \sum_{k=1}^K \pi_k(X_t) m_k(X_t)$ and $\bar{g}(X_t) \equiv \mathbb{E}[g(X_t, S_t) | X_t] = \sum_{k=1}^K \pi_k(X_t) g_k(X_t)$, where $\pi_k(X_t) \equiv P(S_t = k | X_t)$. Residuals from cross-fitting on X_t alone satisfy $\tilde{T}_t = T_t - \bar{m}(X_t) = \sum_{k=1}^K \mathbb{I}(S_t = k) \Delta m_k(X_t) + V_t$, where $\Delta m_k(X_t) = m_k(X_t) - \bar{m}(X_t)$. The outcome residual is:

$$\tilde{Y}_t = Y_t - \mathbb{E}[Y_t | X_t] = \sum_{k=1}^K \theta_k^* \mathbb{I}(S_t = k) T_t - \sum_{k=1}^K \theta_k^* \pi_k(X_t) m_k(X_t) + g(X_t, S_t) - \bar{g}(X_t) + U_t. \quad (8)$$

Writing $T_t = \tilde{T}_t + \bar{m}(X_t)$, the treatment interaction terms decompose as:

$$\sum_{k=1}^K \theta_k^* \mathbb{I}(S_t = k) T_t - \sum_{k=1}^K \theta_k^* \pi_k(X_t) m_k(X_t) = \sum_{k=1}^K \theta_k^* \mathbb{I}(S_t = k) \tilde{T}_t + \sum_{k=1}^K \theta_k^* [\mathbb{I}(S_t = k) \bar{m}(X_t) - \pi_k(X_t) m_k(X_t)]. \quad (9)$$

Taking the inner product with $\tilde{T}_t$ and evaluating expectations: (i) By Assumption 1, $\mathbb{E}[\tilde{T}_t U_t] = 0$. (ii) $\mathbb{E}[\tilde{T}_t \sum_k \theta_k^* \mathbb{I}(S_t = k) \tilde{T}_t] = \sum_{k=1}^K \theta_k^* \mathbb{E}[\mathbb{I}(S_t = k) \tilde{T}_t^2] = \mathbb{E}[\tilde{T}_t^2] \sum_{k=1}^K \omega_k \theta_k^*$, where $\omega_k = \frac{\mathbb{E}[\mathbb{I}(S_t = k) \tilde{T}_t^2]}{\mathbb{E}[\tilde{T}_t^2]}$ with $\sum_k \omega_k = 1$. (iii) Since $\mathbb{E}[\tilde{T}_t | X_t] = 0$ and $\mathbb{E}[\tilde{T}_t \mathbb{I}(S_t = k) | X_t] = \Delta m_k(X_t) \pi_k(X_t)$, the mean adjustment yields $\sum_{k=1}^K \theta_k^* \mathbb{E}[\bar{m}(X_t) \Delta m_k(X_t) \pi_k(X_t)]$. (iv) The confounding component evaluates to $\mathbb{E}[\tilde{T}_t (g(X_t, S_t) - \bar{g}(X_t))] = \sum_{k=1}^K \mathbb{E}[\pi_k(X_t) \Delta m_k(X_t) \Delta g_k(X_t)]$. Dividing by $\mathbb{E}[\tilde{T}_t^2]$ yields (6) and (7). ■

C PROOF OF THEOREM 2

To prove substantive causal non-identification, we constructively exhibit two data-generating distributions, P_1 and P_2 , that are structurally well-formed under the SCM (1)–(2) and satisfy all Assumptions 1–3, yet possess strictly distinct, non-permutation causal effect vectors $\theta^{(1)} \neq \theta^{(2)}$ while inducing identically equal joint distributions over all observable variables:

$$P_1(Y_{1:N}, T_{1:N}, X_{1:N}, Z_{1:N}) \equiv P_2(Y_{1:N}, T_{1:N}, X_{1:N}, Z_{1:N}) \quad \forall N \geq 1. \quad (10)$$

Construction of Substantive Observational Equivalence under Latent Confounding ($T \not\perp S$): Consider a two-state Markov chain ($K = 2$) with symmetric memoryless transition probabilities $P_{01} = P_{10} = 0.5$, yielding independent stationary regimes $S_t \stackrel{\text{i.i.d.}}{\sim} \text{Bernoulli}(0.5)$ with $\pi_0^* = \pi_1^* = 0.5$. Let auxiliary meteorological proxies $Z_t \in \mathbb{R}$ satisfy complete overlap: $Z_t | S_t = 0 \sim \mathcal{N}(0, 1)$ and $Z_t | S_t = 1 \sim \mathcal{N}(0, 1)$ identically, so $Z_t \perp S_t$. Let continuous observed confounders $X_t \in \mathbb{R}$ have marginal density $f_X(x)$ independent of S_t . Let binary treatment assignment $T_t \in \{0, 1\}$ depend directly on the latent regime: $P(T_t = 1 | S_t = 0) = 0.75$ and $P(T_t = 1 | S_t = 1) = 0.25$, so that $T_t \not\perp S_t$, with propensity scores $e(X_t, S_t) \in [0.25, 0.75]$ strictly satisfying Assumption 2 (Overlap & State Positivity with $\eta = 0.25$). Marginally, $P(T_t = 1) = 0.5 \times 0.75 + 0.5 \times 0.25 = 0.5$, and by Bayes' rule, the posterior regime occupancy given treatment satisfies $P(S_t = 0 | T_t = 1) = 0.75$, $P(S_t = 1 | T_t = 1) = 0.25$, $P(S_t = 0 | T_t = 0) = 0.25$, and $P(S_t = 1 | T_t = 0) = 0.75$.

Now consider structural outcome equation $Y_t \in \{0, 1\}$ with $P(Y_t = 1 | T_t = t, S_t = s) = \theta_s t + g_s$ and unconfounded innovations $\mathbb{E}[U_t | X_t, S_t, T_t] = 0$:

1. **Model P_1 (Homogeneous Causal Effect):** Set $\theta^{(1)} = (0.20, 0.20)^\top$ ($\theta_0^{(1)} - \theta_1^{(1)} = 0$), baselines $g_0^{(1)} = 0.30, g_1^{(1)} = 0.50$. Then $P_1(Y_t = 1 | T_t = 0) = 0.25(0.30) + 0.75(0.50) = 0.45$, and $P_1(Y_t = 1 | T_t = 1) = 0.75(0.20 + 0.30) + 0.25(0.20 + 0.50) = 0.75(0.50) + 0.25(0.70) = 0.55$.
2. **Model P_2 (Heterogeneous Causal Effect):** Set $\theta^{(2)} = (0.10, -0.30)^\top$ ($\theta_0^{(2)} - \theta_1^{(2)} = +0.40$), baselines $g_0^{(2)} = 0.60, g_1^{(2)} = 0.40$. Then $P_2(Y_t = 1 | T_t = 0) = 0.25(0.60) + 0.75(0.40) = 0.45$, and $P_2(Y_t = 1 | T_t = 1) = 0.75(0.10 + 0.60) + 0.25(-0.30 + 0.40) = 0.75(0.70) + 0.25(0.10) = 0.525 + 0.025 = 0.55$.

Because S_t transitions independently ($p = 0.5$), observations (Y_t, T_t, X_t, Z_t) are mutually independent across time, yielding $P(Y_{1:N}, T_{1:N}, X_{1:N}, Z_{1:N}) = \prod_{t=1}^N P(Y_t, T_t, X_t, Z_t)$. Comparing single-step observable conditionals, $P_1(Y_t | T_t, X_t, Z_t) \equiv P_2(Y_t | T_t, X_t, Z_t)$ identically for all (y, t, x, z) , ensuring $P_1(Y_{1:N}, T_{1:N}, X_{1:N}, Z_{1:N}) \equiv P_2(Y_{1:N}, T_{1:N}, X_{1:N}, Z_{1:N})$ for all $N \geq 1$. Yet structural effects satisfy $\theta^{(1)} = (0.20, 0.20)^\top \neq (0.10, -0.30)^\top = \theta^{(2)}$,

which is not a permutation. **Dynamic Identifiability:** When regimes exhibit non-trivial temporal dependence ($p \neq 0.5$), latent Markov models can achieve generic algebraic point-identification from sequence transitions of length $N \geq 3$ (Allman et al., 2009). However, in memoryless/cross-sectional regimes, or whenever proxy uncertainty leaves states unconstrained without dynamic modeling, structural non-identification holds identically for all $N \geq 1$. ■

D PROOF OF THEOREM 3

This appendix proves Theorem 3 for posterior-weighted nuisances; Appendix E adapts the argument to posterior-adjusted nuisances. **Full statement.** Let $\mathbf{H}_t = \mathbf{e}_{S_t}$, $\gamma_t \in \Delta^K$ a state proxy with $\varepsilon_\gamma \equiv \mathbb{E}\|\gamma_t - \mathbf{H}_t\|_1$, and $\mathbf{J}, \mathbf{S}$ the population coupled Gram matrix and score at the proxy nuisances $(m_k^\gamma, \mu_k^\gamma)$. Under Assumption 2 and $\|\boldsymbol{\theta}^*\|_2 \leq M_\theta$, if $\lambda_{\min}(\mathbf{J}) > 0$: (i) $\boldsymbol{\theta}_\gamma \equiv \mathbf{J}^{-1}\mathbf{S}$ satisfies $\|\boldsymbol{\theta}_\gamma - \boldsymbol{\theta}^*\|_2 \leq C\varepsilon_\gamma/\lambda_{\min}(\mathbf{J})$ with $C = C_S + M_\theta C_J$, $C_S = \sqrt{K}(M_T M_Y + M_Y \tilde{C}_m + M_T \tilde{C}_\mu)$, $C_J = K(2M_T^2 + 2M_T \tilde{C}_m)$, $\tilde{C}_m = 2B_T/c_0^2$ and $\tilde{C}_\mu = 2B_Y/c_0^2$. (ii) With $r_N = \max_k(\|\hat{m}_k - m_k^\gamma\|_{L_2} + \|\hat{\mu}_k - \mu_k^\gamma\|_{L_2})$, under purged cross-fitting and Assumption 4, on $\mathcal{E}_N = \{\|\hat{\mathbf{J}} - \mathbf{J}\|_{\text{op}} \leq \lambda_{\min}(\mathbf{J})/2\}$, $\|\hat{\boldsymbol{\theta}}_\gamma - \boldsymbol{\theta}^*\|_2 \leq C\varepsilon_\gamma/\lambda_{\min}(\mathbf{J}) + \mathcal{O}_P((N^{-1/2} + r_N)/\lambda_{\min}(\mathbf{J}))$.

Let $\mathbf{H}_t = \mathbf{e}_{S_t} \in \{0, 1\}^K$ denote the unobserved true regime indicator vector. Under SCM (1)–(2), the oracle regime nuisance functions are:

$$m_k^*(X_t) \equiv \mathbb{E}[T_t | X_t, S_t = k], \quad \mu_k^*(X_t) \equiv \mathbb{E}[Y_t | X_t, S_t = k] = \theta_k^* m_k^*(X_t) + g(X_t, k). \quad (11)$$

Step 1: Define proxy-weighted targets and population estimand. The empirical cross-fitted learners $\hat{m}_k$ and $\hat{\mu}_k$ minimize the posterior-weighted empirical risk targeting the proxy-weighted conditional expectations:

$$m_k^\gamma(X_t) \equiv \frac{\mathbb{E}[\gamma_{tk} T_t | X_t]}{\mathbb{E}[\gamma_{tk} | X_t]}, \quad \mu_k^\gamma(X_t) \equiv \frac{\mathbb{E}[\gamma_{tk} Y_t | X_t]}{\mathbb{E}[\gamma_{tk} | X_t]}. \quad (12)$$

Define the centered residuals at proxy targets: $\tilde{T}_{tk}^\gamma \equiv T_t - m_k^\gamma(X_t)$, $\tilde{Y}_{tk}^\gamma \equiv Y_t - \mu_k^\gamma(X_t)$, and at oracle targets: $\tilde{T}_{tk}^* \equiv T_t - m_k^*(X_t)$, $\tilde{Y}_{tk}^* \equiv Y_t - \mu_k^*(X_t)$. The population proxy Gram matrix $\mathbf{J} \in \mathbb{R}^{K \times K}$ and score vector $\mathbf{S} \in \mathbb{R}^K$ are:

$$J_{jk} \equiv \mathbb{E}[\gamma_{tj} \gamma_{tk} \tilde{T}_{tj}^\gamma \tilde{T}_{tk}^\gamma], \quad S_k \equiv \mathbb{E}[\gamma_{tk} \tilde{T}_{tk}^\gamma \tilde{Y}_{tk}^\gamma], \quad (13)$$

and their oracle counterparts $\mathbf{J}^*, \mathbf{S}^*$ are defined by:

$$J_{jk}^* \equiv \mathbb{E}[H_{tj} H_{tk} \tilde{T}_{tj}^* \tilde{T}_{tk}^*] = \delta_{jk} \mathbb{E}[H_{tk} (\tilde{T}_{tk}^*)^2], \quad S_k^* \equiv \mathbb{E}[H_{tk} \tilde{T}_{tk}^* \tilde{Y}_{tk}^*], \quad (14)$$

where $H_{tj} H_{tk} = \delta_{jk} H_{tk}$ because latent regimes are mutually exclusive.

Step 2: Orthogonality structure of the coupled score (Lemma 2).

Lemma 2 (Orthogonality Structure of the Coupled Score). *Let $\Psi_k(\boldsymbol{\theta}; \eta) = \gamma_k \tilde{T}_k (\tilde{Y}_k - \sum_j \gamma_j \tilde{T}_j \theta_j)$ with $\tilde{T}_k = T - m_k(X)$, $\tilde{Y}_k = Y - \mu_k(X)$, so $\tilde{\mathbf{S}} - \hat{\mathbf{J}}\boldsymbol{\theta} = \frac{1}{N} \sum_t \boldsymbol{\Psi}_t$. At $\eta^\gamma = (m^\gamma, \mu^\gamma)$ and any $\boldsymbol{\theta}$, with $g(X) \equiv \mathbb{E}[\|\gamma - \mathbf{H}\|_1 | X]$:*

- (i) $\partial_\mu \mathbb{E} \Psi_k[\Delta \mu] = 0$ for any proxy γ .
- (ii) The cross-moment part $\gamma_k \tilde{T}_k \tilde{Y}_k$ is orthogonal in both directions.
- (iii) $|\partial_m \mathbb{E} \Psi_k[\Delta m]| \leq 2M_T \|\boldsymbol{\theta}\|_1 \sqrt{2\varepsilon_\gamma} \max_i \|\Delta m_i\|_{L_2}$. For hard assignments $\gamma_t \in \{0, 1\}^K$, this derivative is identically 0.

Proof. (i)–(ii): $\mathbb{E}[\gamma_k (T - m_k^\gamma) | X] = 0$ and $\mathbb{E}[\gamma_k (Y - \mu_k^\gamma) | X] = 0$ by definition of m_k^γ, μ_k^γ . (iii): differentiating $-\gamma_k \tilde{T}_k \sum_j \gamma_j \tilde{T}_j \theta_j$ in direction Δm gives $\sum_j \theta_j \mathbb{E}[\Delta m_k \gamma_k \gamma_j \tilde{T}_j + \Delta m_j \gamma_k \gamma_j \tilde{T}_k]$. For $j = k$, $\mathbb{E}[\gamma_k^2 \tilde{T}_k | X] = \mathbb{E}[(\gamma_k^2 - \gamma_k) \tilde{T}_k | X]$ since $\mathbb{E}[\gamma_k \tilde{T}_k | X] = 0$, bounded by $M_T |\gamma_k - H_k|$. For $j \neq k$, $\gamma_k \gamma_j \leq |\gamma_k - H_k| + |\gamma_j - H_j|$. Bounding each moment by $M_T g(X)$ and using $\|g\|_{L_2} \leq \sqrt{2\varepsilon_\gamma}$ yields (iii). For hard assignments, $\gamma_k \gamma_j = \delta_{kj} \gamma_k$ so both moments vanish. □

Step 3: Isolate proxy-induced bias via oracle score identity. Under SCM (1)–(2), on the event $S_t = k$ ($H_{tk} = 1$), the oracle outcome residual is $\tilde{Y}_{tk}^* = Y_t - \mu_k^*(X_t) = \theta_k^* \tilde{T}_{tk}^* + U_t$. Substituting into S_k^* :

$$S_k^* = \mathbb{E} \left[H_{tk} \tilde{T}_{tk}^* (\theta_k^* \tilde{T}_{tk}^* + U_t) \right] = \theta_k^* \mathbb{E}[H_{tk} (\tilde{T}_{tk}^*)^2] + \mathbb{E} \left[H_{tk} \tilde{T}_{tk}^* \mathbb{E}[U_t | X_t, S_t, T_t, Z_{1:N}] \right] = \theta_k^* J_{kk}^*, \quad (15)$$

where the innovation term vanishes by Assumption 1. Because $\mathbf{J}^* = \text{diag}(J_{11}^*, \dots, J_{KK}^*)$, this yields the exact vector identity:

$$\mathbf{S}^* = \mathbf{J}^* \boldsymbol{\theta}^*. \quad (16)$$

Now consider the proxy score discrepancy $\mathbf{b}_\gamma \equiv \mathbf{S} - \mathbf{J}\boldsymbol{\theta}^*$. Expanding around the oracle identity:

$$\mathbf{b}_\gamma = (\mathbf{S} - \mathbf{S}^*) - (\mathbf{J} - \mathbf{J}^*)\boldsymbol{\theta}^*. \quad (17)$$

Step 4: Bound score and Gram proxy discrepancies. We bound each component via telescoping expansions in $\varepsilon_\gamma \equiv \mathbb{E}\|\boldsymbol{\gamma}_t - \mathbf{H}_t\|_1$. First, under positivity (Assumption 2), $\pi_k(x) \equiv \mathbb{E}[H_{tk} \mid X_t = x] \geq c_0 > 0$ and $\pi_k^\gamma(x) \equiv \mathbb{E}[\gamma_{tk} \mid X_t = x] \geq c_0 > 0$. For any bounded variable V_t with $|V_t| \leq M_V$ almost surely:

$$\begin{aligned} \left| \frac{\mathbb{E}[\gamma_{tk} V_t \mid X_t]}{\pi_k^\gamma(X_t)} - \frac{\mathbb{E}[H_{tk} V_t \mid X_t]}{\pi_k(X_t)} \right| &\leq \frac{\mathbb{E}[|\gamma_{tk} - H_{tk}| |V_t| \mid X_t]}{\pi_k^\gamma(X_t)} + \mathbb{E}[H_{tk} |V_t| \mid X_t] \left| \frac{\pi_k(X_t) - \pi_k^\gamma(X_t)}{\pi_k^\gamma(X_t) \pi_k(X_t)} \right| \\ &\leq \frac{2M_V}{c_0^2} \mathbb{E}[|\gamma_{tk} - H_{tk}| \mid X_t]. \end{aligned} \quad (18)$$

Applying this bound to $V_t = T_t$ and $V_t = Y_t$ yields $\mathbb{E}|m_k^\gamma(X_t) - m_k^*(X_t)| \leq \tilde{C}_m \varepsilon_\gamma$ and $\mathbb{E}|\mu_k^\gamma(X_t) - \mu_k^*(X_t)| \leq \tilde{C}_\mu \varepsilon_\gamma$. Second, expanding the score difference $S_k - S_k^* = \mathbb{E}[\gamma_{tk} \tilde{T}_{tk}^\gamma \tilde{Y}_{tk}^\gamma - H_{tk} \tilde{T}_{tk}^* \tilde{Y}_{tk}^*]$:

$$\gamma_{tk} \tilde{T}_{tk}^\gamma \tilde{Y}_{tk}^\gamma - H_{tk} \tilde{T}_{tk}^* \tilde{Y}_{tk}^* = (\gamma_{tk} - H_{tk}) \tilde{T}_{tk}^\gamma \tilde{Y}_{tk}^\gamma - H_{tk} (m_k^\gamma - m_k^*) \tilde{Y}_{tk}^\gamma - H_{tk} \tilde{T}_{tk}^* (\mu_k^\gamma - \mu_k^*). \quad (19)$$

Under bounded support ($\sup_t |\tilde{T}_{tk}^\gamma| \leq M_T$, $\sup_t |\tilde{Y}_{tk}^\gamma| \leq M_Y$), taking expectations yields $|S_k - S_k^*| \leq (M_T M_Y + M_Y \tilde{C}_m + M_T \tilde{C}_\mu) \varepsilon_\gamma \equiv C_S \varepsilon_\gamma$. Third, expanding the Gram difference $J_{jk} - J_{jk}^* = \mathbb{E}[\gamma_{tj} \gamma_{tk} \tilde{T}_{tj}^\gamma \tilde{T}_{tk}^\gamma - H_{tj} H_{tk} \tilde{T}_{tj}^* \tilde{T}_{tk}^*]$:

$$\begin{aligned} \gamma_{tj} \gamma_{tk} \tilde{T}_{tj}^\gamma \tilde{T}_{tk}^\gamma - H_{tj} H_{tk} \tilde{T}_{tj}^* \tilde{T}_{tk}^* &= (\gamma_{tj} \gamma_{tk} - H_{tj} H_{tk}) \tilde{T}_{tj}^\gamma \tilde{T}_{tk}^\gamma - H_{tj} H_{tk} (m_j^\gamma - m_j^*) \tilde{T}_{tk}^\gamma \\ &\quad - H_{tj} H_{tk} \tilde{T}_{tj}^* (m_k^\gamma - m_k^*). \end{aligned} \quad (20)$$

Since $|\gamma_{tj} \gamma_{tk} - H_{tj} H_{tk}| \leq |\gamma_{tj} - H_{tj}| + |\gamma_{tk} - H_{tk}|$, taking expectations gives $|J_{jk} - J_{jk}^*| \leq (2M_T^2 + 2M_T \tilde{C}_m) \varepsilon_\gamma$, so $\|\mathbf{J} - \mathbf{J}^*\|_{\text{op}} \leq C_J \varepsilon_\gamma$. Combining bounds yields:

$$\|\mathbf{b}_\gamma\|_2 \leq \|\mathbf{S} - \mathbf{S}^*\|_2 + \|\mathbf{J} - \mathbf{J}^*\|_{\text{op}} \|\boldsymbol{\theta}^*\|_2 \leq (C_S + C_J M_\theta) \varepsilon_\gamma \equiv C \cdot \varepsilon_\gamma. \quad (21)$$

Step 5: Population bound and finite-sample rate. Since $\mathbf{S} = \mathbf{J}\boldsymbol{\theta}^* + \mathbf{b}_\gamma$ and $\boldsymbol{\theta}_\gamma \equiv \mathbf{J}^{-1} \mathbf{S}$, we have $\boldsymbol{\theta}_\gamma - \boldsymbol{\theta}^* = \mathbf{J}^{-1} \mathbf{b}_\gamma$, so $\|\boldsymbol{\theta}_\gamma - \boldsymbol{\theta}^*\|_2 \leq \frac{\|\mathbf{b}_\gamma\|_2}{\lambda_{\min}(\mathbf{J})} \leq \frac{C \varepsilon_\gamma}{\lambda_{\min}(\mathbf{J})}$, establishing part (i). For part (ii): since $\mathbf{J}\boldsymbol{\theta}_\gamma = \mathbf{S}$, $\hat{\boldsymbol{\theta}}_\gamma - \boldsymbol{\theta}_\gamma = \hat{\mathbf{J}}^{-1}[(\hat{\mathbf{S}} - \mathbf{S}) - (\hat{\mathbf{J}} - \mathbf{J})\boldsymbol{\theta}_\gamma]$. By Weyl's inequality, $\lambda_{\min}(\hat{\mathbf{J}}) \geq \lambda_{\min}(\mathbf{J}) - \|\hat{\mathbf{J}} - \mathbf{J}\|_{\text{op}} \geq \lambda_{\min}(\mathbf{J})/2$ on event $\mathcal{E}_N$, so $\|\hat{\mathbf{J}}^{-1}\|_{\text{op}} \leq 2/\lambda_{\min}(\mathbf{J})$. Lemma 2 gives $\|\hat{\mathbf{S}} - \mathbf{S}\|_2 = \mathcal{O}_P(N^{-1/2} + r_N^2)$, while non-orthogonality of the coupled Gram matrix yields $\|\hat{\mathbf{J}} - \mathbf{J}\|_{\text{op}} = \mathcal{O}_P(N^{-1/2} + r_N)$. Combining with $\|\boldsymbol{\theta}_\gamma\|_2 \leq M_\theta + C \varepsilon_\gamma / \lambda_{\min}(\mathbf{J})$ establishes the finite-sample bound in (ii). ■

Proposition 2 (Exact Proxy Bias in the Symmetric Flip Model). *Let $K = 2$, $\boldsymbol{\pi}^* = (\frac{1}{2}, \frac{1}{2})$, $T_t = V_t$ with $V_t \perp (S_t, X_t, U_t)$, $\mathbb{E}V_t = 0$, $\text{Var}V_t = \sigma^2$, $Y_t = \theta_{S_t} T_t + g_{S_t} + U_t$, $\mathbb{E}[U_t \mid T_t, S_t] = 0$, and $\boldsymbol{\gamma}_t = (1 - \frac{\varepsilon}{2})\mathbf{H}_t + \frac{\varepsilon}{2}(\mathbf{1} - \mathbf{H}_t)$ with $\varepsilon \in [0, 1)$ ($\varepsilon_\gamma = \varepsilon$). Then $\lambda_{\min}(\mathbf{J}) = \sigma^2(1 - \varepsilon)^2/2$ and*

$$\boldsymbol{\theta}_\gamma - \boldsymbol{\theta}^* = \frac{\varepsilon}{1 - \varepsilon} \cdot \frac{\theta_1 - \theta_2}{2} (1, -1)^\top, \quad \|\boldsymbol{\theta}_\gamma - \boldsymbol{\theta}^*\|_2 = \frac{\varepsilon}{1 - \varepsilon} \frac{|\theta_1 - \theta_2|}{\sqrt{2}}. \quad (22)$$

Proof. $m_k^\gamma = 0$ since $\mathbb{E}V = 0$, $V \perp S$, so $\tilde{T}_k = V$ and constants in $\tilde{Y}_k$ drop out of $\mathbf{S}$. With $a = 1 - \varepsilon/2$, $b = \varepsilon/2$, $\mathbf{M} = \begin{pmatrix} a & b \\ b & a \end{pmatrix}$: $\mathbf{J} = \frac{\sigma^2}{2} \mathbf{M}^2$ and $\mathbf{S} = \frac{\sigma^2}{2} \mathbf{M} \boldsymbol{\theta}^*$. Since $\mathbf{M}$ has eigenvalues 1 (eigenvector $(1, 1)^\top$) and $1 - \varepsilon$ (eigenvector $(1, -1)^\top$), $\boldsymbol{\theta}_\gamma = \mathbf{J}^{-1} \mathbf{S} = \mathbf{M}^{-1} \boldsymbol{\theta}^*$, $\lambda_{\min}(\mathbf{J}) = \sigma^2(1 - \varepsilon)^2/2$, and $(\mathbf{M}^{-1} - \mathbf{I})\boldsymbol{\theta}^*$ evaluates to the displayed vector. □

Remark on Proxy Construction and Scope of Lower Bound. In Proposition 2, the proxy $\boldsymbol{\gamma}$ is constructed directly as a corrupted function of the latent state $\mathbf{H}$ rather than inferred from observable features Z . An estimator having access to $\boldsymbol{\gamma}$ directly could in principle invert the corruption matrix to recover $\mathbf{H}$ consistently. Thus, Proposition 2 shows that the plug-in bound $\mathcal{O}(\varepsilon_\gamma / \lambda_{\min}(\mathbf{J}))$ is attained by a corrupted proxy, but does not assert that calibrated posteriors inferred from Z necessarily incur this bias, nor do we make any minimax lower bound claim. The factorial experiment's synthetic proxy manipulation ($\rho = 0.995$) shares this design.

E POSTERIOR-ADJUSTED NUISANCES

Throughout, $F_t = (X_t, \gamma_t, \gamma_t \otimes X_t)$, $m^\gamma(F) = \mathbb{E}[T | F]$, $\mu^\gamma(F) = \mathbb{E}[Y | F]$, $\tilde{T} = T - m^\gamma$, $\tilde{Y} = Y - \mu^\gamma$, and the population system is $J_{jk} = \mathbb{E}[\gamma_j \gamma_k \tilde{T}^2]$, $S_k = \mathbb{E}[\gamma_k \tilde{T} \tilde{Y}]$.

Lemma 1 in detail. For a direction $\delta_m(F)$, $\partial_t \mathbb{E} \psi_k(\theta; m^\gamma + t \delta_m, \mu^\gamma)|_{t=0} = -\mathbb{E}[\gamma_k \delta_m \{\tilde{Y} - \sum_j \gamma_j \theta_j \tilde{T}\}] + \mathbb{E}[\gamma_k \tilde{T} \sum_j \gamma_j \theta_j \delta_m]$. Conditioning on F , the first term is $-\mathbb{E}[\gamma_k \delta_m (\mathbb{E}[\tilde{Y} | F] - \sum_j \gamma_j \theta_j \mathbb{E}[\tilde{T} | F])] = 0$, and the second is $\mathbb{E}[\gamma_k \sum_j \gamma_j \theta_j \delta_m \mathbb{E}[\tilde{T} | F]] = 0$. The μ -direction gives $-\mathbb{E}[\gamma_k \delta_\mu \mathbb{E}[\tilde{T} | F]] = 0$, and $\partial_t \mathbb{E}[\gamma_j \gamma_k (\tilde{T} - t \delta_m)^2] = -2\mathbb{E}[\gamma_j \gamma_k \delta_m \mathbb{E}[\tilde{T} | F]] = 0$. Nothing here uses calibration or the form of γ : the weights only need to be functions of the conditioning set. For posterior-weighted regime nuisances $m_k(X) = \mathbb{E}[\gamma_k T | X] / \mathbb{E}[\gamma_k | X]$, the same calculation leaves $\mathbb{E}[\gamma_k \gamma_j \delta_m(X) \tilde{T}_k] \neq 0$ for soft weights, the $\sqrt{\varepsilon_\gamma}$ sensitivity of Appendix D. *Numerical check* (`src/synthesis/coupled_law_check.py`, $N = 10^6$): perturbing the treatment nuisance by $t(1 + X^2 + \gamma)$, the finite-difference derivative of the mean score is 0.4% of the Jacobian norm for posterior-adjusted nuisances (sampling noise) and 85% for posterior-weighted nuisances.

Theorem 3 for posterior-adjusted nuisances. Assume (i) calibration given F : $P(S_t = k | F_t) = \gamma_{tk}$, and (ii) $(T, Y) \perp Z | (X, S)$, so that $\mathbb{E}[T | X, S, \gamma] = m_S(X)$. Then $m^\gamma(F) = \sum_k \gamma_k m_k(X)$ and $|m^\gamma - m_S| \leq B_T \|\gamma - \mathbf{H}\|_1$, and likewise for μ^γ . Steps 3–5 of Appendix D then go through with the shared residual $\tilde{T}$ in place of $\tilde{T}_k$ and $(\tilde{C}_m, \tilde{C}_\mu) = (B_T, B_Y)$, giving $\|\mathbf{b}_\gamma\|_2 \leq C\varepsilon_\gamma$. For the stochastic part, Lemma 1 makes both $\hat{\mathbf{S}}$ and $\hat{\mathbf{J}}$ first-order insensitive to nuisance error, so $\|\hat{\mathbf{S}} - \mathbf{S}\| + \|\hat{\mathbf{J}} - \mathbf{J}\|_{\text{op}} = \mathcal{O}_P(N^{-1/2} + r_N^2)$ under purged cross-fitting, and the argument of Step 5 gives the stated rate with $\rho_N = r_N^2$. When (i) fails, the bound acquires the calibration error $\mathbb{E}\|P(S \in \cdot | F) - \gamma\|_1$. $\square$

Proof of Theorem 4. Write $r_t = S_t - \gamma_t$; by calibration $\mathbb{E}[r | F] = 0$ and $\mathbb{E}[r^2 | F] = v$. Then $\mathbb{E}[T | F] = m(X) + a\gamma$ and $\tilde{T} = ar + V$. Since $\theta_S = \theta_0 + \Delta\theta S$ with $\Delta\theta = \theta_1 - \theta_0$ and $S^2 = S$, $\mathbb{E}[\theta_S T | F] = \theta_0(m + a\gamma) + \Delta\theta\gamma(m + a)$, so

$$\tilde{Y} = \theta_0 \tilde{T} + \Delta\theta \{(S - \gamma)(m + a) + SV\} + br + U.$$

For OR-DML, $\mathbf{w}^\top \theta^* = \theta_0 + \gamma \Delta\theta$, and the residual at the truth is $e = \tilde{Y} - (\theta_0 + \gamma \Delta\theta) \tilde{T} = r\{\Delta\theta(m + a(1 - \gamma)) + b\} + \Delta\theta rV + U = cr + \Delta\theta rV + U$. Expanding $\tilde{T}e = (ar + V)(cr + \Delta\theta rV + U)$ and conditioning on F (with V independent of (S, F) , $\mathbb{E}[U | S, F, V] = 0$) leaves only acv . Hence $\mathbf{b}_\gamma = \mathbb{E}[\mathbf{w} \tilde{T} e] = a\mathbb{E}[\mathbf{w} cv]$ and, from $\mathbb{E}[\tilde{T}^2 | F] = a^2v + \sigma^2$, $\mathbf{J} = \mathbb{E}[\mathbf{w} \mathbf{w}^\top (a^2v + \sigma^2)]$. For the pooled estimator ($w \equiv 1$), $\mathbb{E}[\tilde{T} \tilde{Y}] = \mathbb{E}[(\theta_0 + \gamma \Delta\theta)(a^2v + \sigma^2)] + a\mathbb{E}[cv]$, and dividing by $\mathbb{E}[\tilde{T}^2]$ gives the stated limit. Consistency of the sample version follows as in Appendix G, Steps 1–4, with remainder $\mathcal{O}_P(N^{-1/2} + r_N^2 + \varepsilon_\tau)$ by Lemma 1. (a) is immediate. (b) is Appendix G. (c): $\mathbf{J} - \mathbb{E}[\mathbf{w} \mathbf{w}^\top] \sigma^2 = a^2 \mathbb{E}[\mathbf{w} \mathbf{w}^\top v] \succeq 0$ grows in $|a|$ and in v . $\square$

Numerical check. Table 4 compares the OR-DML estimate with the law on $N = 10^6$ draws with a calibrated Beta posterior, $m(X)$ linear and nuisances fitted on $(1, X, \gamma, \gamma X)$. The bias is hump-shaped in a (largest at $a = 4$) while $\lambda_{\min}(\mathbf{J})$ rises from 0.23 to 1.27, past the oracle value 0.50.

Table 4: Residual-state law for posterior-adjusted OR-DML: estimate versus law, true $\theta^* = (0.5, 0.5 + \Delta\theta)$ (`reports/synthesis/coupled_law_check.csv`).

a	Δ_g	$\Delta\theta$	σ	estimate	law	dev.	$\lambda_{\min}(\mathbf{J})$
0	3	1	1	(0.505, 1.500)	(0.500, 1.500)	0.0048	0.227
0.5	3	1	1	(0.725, 1.708)	(0.723, 1.705)	0.0029	0.231
1	3	1	1	(0.961, 1.882)	(0.953, 1.886)	0.0085	0.244
2	3	1	1	(1.317, 2.094)	(1.317, 2.095)	0.0008	0.292
4	3	1	1	(1.625, 2.091)	(1.624, 2.090)	0.0010	0.487
8	3	1	1	(1.697, 1.874)	(1.695, 1.875)	0.0013	1.266
4	-2	0.5	0.7	(0.461, 0.648)	(0.462, 0.646)	0.0018	0.370
1.5	3	0	1	(0.967, 0.967)	(0.969, 0.970)	0.0026	0.264
1	3	1	1	(1.141, 2.019)	(1.146, 2.021)	0.0048	0.114

F THE COUPLED SCORE UNDER STATE CONFUSION

F.1 Proof of Proposition 1

With oracle regime-wise nuisances $m_k(X) = \mathbb{E}[T \mid X, S=k]$ and $\mu_k(X) = \mathbb{E}[Y \mid X, S=k]$, the regime- k residuals evaluated on an observation from regime s are $\tilde{T}_k = (m_s - m_k) + \sigma_s V$ and $\tilde{Y}_k = (\mu_s - \mu_k) + \theta_s \sigma_s V + U$, where V has mean zero and unit variance and is independent of U and of S . The covariate parts cancel because the nuisances are regime-correct; only the regime-mean contrasts remain. Because γ_j is a function of s alone,

$$J_{jk} = \mathbb{E}[\gamma_j \gamma_k \tilde{T}_j \tilde{T}_k] = \sum_s \pi_s \gamma_j(s) \gamma_k(s) \mathbb{E}[\tilde{T}_j \tilde{T}_k \mid S=s] = \sum_s \pi_s \gamma_j(s) \gamma_k(s) [(m_s - m_j)(m_s - m_k) + \sigma_s^2],$$

and $S_j = \mathbb{E}[\gamma_j \tilde{T}_j \tilde{Y}_j] = \sum_s \pi_s \gamma_j(s) [(m_s - m_j)(\mu_s - \mu_j) + \theta_s \sigma_s^2]$. In the symmetric case put $a = 1 - \alpha$, $b = \alpha$. The diagonal entries are $\frac{1}{2}[a^2 \sigma^2 + b^2(\Delta^2 + \sigma^2)]$; the off-diagonal entry is $\frac{1}{2} \cdot 2ab\sigma^2 = ab\sigma^2$, because on either regime the two residuals are σV and $\pm\Delta + \sigma V$, whose product has mean σ^2 . A symmetric 2×2 matrix with equal diagonal has eigenvectors $(1, \pm 1)$ and eigenvalues “diagonal $\pm$ off-diagonal”, which gives $\lambda_{\pm}$ after using $a + b = 1$. For the displacement, write $\mathbf{S} - \mathbf{J}\boldsymbol{\theta}^* = \mathbf{b}_\gamma$, project $\mathbf{b}_\gamma$ on $(1, \pm 1)$ and divide by $2\lambda_{\pm}$. The $(1, 1)$ component of $\mathbf{b}_\gamma$ collects the leaked outcome contrast $\alpha\Delta G$ and the quadratic leakage $-\frac{1}{2}\alpha^2\Delta^2(\theta_0 + \theta_1)$. The $(1, -1)$ component collects the effect heterogeneity. The identity is verified symbolically in `src/synthesis/closed_form_factorial.py` (`displacement_formula_ok`). Setting $\Delta = 0$ recovers the symmetric flip model of Proposition 2. $\square$

Proof of Corollary 1. (a) $\lambda_-(\varepsilon_\gamma) - \sigma^2/2 = \frac{1}{2}\sigma^2(\varepsilon_\gamma^2 - 2\varepsilon_\gamma) + \varepsilon_\gamma^2\Delta^2/8$, which is positive iff $\varepsilon_\gamma(\sigma^2 + \Delta^2/4) > 2\sigma^2$. (b) Dropping the $\mathcal{O}(\alpha^2)$ numerator term and the heterogeneity component, $\|\boldsymbol{\theta}_\gamma - \boldsymbol{\theta}^*\|_2 \approx \sqrt{2}\alpha|\Delta G|/(\sigma^2 + \alpha^2\Delta^2)$ with $\alpha = \varepsilon_\gamma/2$. Maximizing $\varepsilon_\gamma/(\sigma^2 + \varepsilon_\gamma^2\Delta^2/4)$ gives $\varepsilon^* = 2\sigma/|\Delta|$ (also verified symbolically) and the stated peak. (c) The leading displacement is proportional to $(1, 1)$, the eigenvector of λ_+ , so $\|\mathbf{J}^{-1}\mathbf{b}_\gamma\| = \|\mathbf{b}_\gamma\|/\lambda_+ \leq \|\mathbf{b}_\gamma\|/\lambda_-$. $\square$

F.2 Verification against the factorial experiment

Table 5 reports, for every cell of the 7×7 factorial with $\varepsilon_\gamma > 0$, the ratio of the observed median error (100 replications) to the population error from Proposition 1, using the factorial’s own parameters: stationary weights $(0.556, 0.444)$, $m = (2, 5)$, $g = (15, 50)$, $\theta = (0.75, 2.5)$, $\sigma_0 = 1$. All 42 ratios lie in $[0.98, 1.02]$, and the population $\lambda_{\min}$ matches the mean realized value to within 1.1%. The finite-sample component is visible only at $\varepsilon_\gamma = 0$, where the population error is zero.

Table 5: Observed median error divided by the exact population error (Proposition 1), factorial design.

$\sigma_1 \backslash \varepsilon_\gamma$	0.01	0.02	0.05	0.1	0.2	0.4
0.08	0.984	0.986	0.998	0.998	0.999	1.001
0.15	0.987	0.985	0.989	1.000	0.999	1.000
0.3	0.992	0.989	0.987	0.990	1.000	1.000
0.5	1.000	0.994	0.991	0.990	0.995	0.999
0.75	1.006	0.997	0.995	0.994	0.996	1.000
1	1.008	1.014	1.017	1.011	1.007	1.009
1.5	1.014	1.012	1.018	1.021	1.020	1.022

F.3 Phantom resolution beyond the coupled score

Hard classifiers. Let $J_{\text{true}} = \mathbb{E}[\tilde{T}_t^2 H_t H_t^\top]$ with true residual treatment variances $a > b > 0$ in a balanced two-state model. A symmetric hard classifier that flips labels with probability $p \in (0, 1/2)$ yields inferred diagonal information proportional to $((1-p)a + pb, pa + (1-p)b)/2$. The smaller inferred value $[pa + (1-p)b]/2$ exceeds $b/2$, and at $p = 1/2$ the apparent condition number is one however large the oracle condition number a/b is. In a three-state Monte Carlo with residual treatment standard deviations $(1, 0.3, 0.08)$ and $\Delta_Z = 0.5$, oracle $\lambda_{\min} \approx 0.0022$, while the soft and hard generated-state values are about 0.0246 and 0.0457 (11.4 $\times$ and 21.2 $\times$ inflation; research ledger in the supplementary archive).

Why posterior means are insufficient for information moments. For one-hot H_t and posterior $q_t = P(S_t \mid \mathcal{I}_t)$, $\mathbb{E}[H_t H_t^\top \mid \mathcal{I}_t] = \text{diag}(q_t)$, not $q_t q_t^\top$. For any weight W_t measurable with respect to $\mathcal{I}_t$, $\mathbb{E}[W_t H_t H_t^\top] =$

$\mathbb{E}[W_t \text{diag}(q_t)]$. This can recover oracle second moments under a correctly specified posterior whose information set contains every state-informative variable entering W_t . It is not a free correction, though: misspecifying the state-treatment model recreates phantom resolution, so we treat completed geometry as a sensitivity diagnostic.

Directional reliability. The bound $\|\mathbf{b}_\gamma\|/\lambda_{\min}(\mathbf{J})$ is worst-directional. For a scientific contrast $\mathbf{c}$ the relevant first-order quantity is $|\mathbf{c}^\top \mathbf{J}^{-1} \mathbf{b}_\gamma|$. In $K \geq 3$ regimes, equal-magnitude perturbations can therefore have very different consequences depending on their alignment with the information eigenvectors, and no task-agnostic scalar representation score is sufficient in general. Fine latent states may still be needed for nuisance adjustment even when a separate coefficient for every state is unsupported, which is why we distinguish adjustment resolution from target resolution.

G THE RESIDUAL-STATE BIAS LAW

Setting. $S_t \in \{0, \dots, K-1\}$ with indicator vector $D_t \in \{0, 1\}^{K-1}$; $T_t = m(X_t) + a^\top D_t + V_t$ and $Y_t = \theta T_t + g(X_t) + b^\top D_t + U_t$; $\hat{\gamma}_t = \hat{\gamma}_t(Z_{1:N})$ is any posterior computed from the proxies; nuisances are cross-fitted on $F_t = (X_t, \hat{\gamma}_t)$ with purged blocks; $\hat{\theta} = \sum \tilde{T}_t \tilde{Y}_t / \sum \tilde{T}_t^2$.

Assumption 5 (Exogenous noise). V_t is independent of (S, Z, X) at all times, $\mathbb{E}V_t = 0$, $\text{Var}V_t = \sigma^2 > 0$; $\mathbb{E}[U_t | S, Z, X, V] = 0$; both noises have summable autocovariances and bounded $4+\delta$ moments.

Assumption 6 (Posterior as a local feature). The process is stationary and α -mixing; $\hat{\gamma}_t$ depends on Z_s for $|s - t| > \tau$ only through terms of size ε_τ in L^2 (filter stability).

Assumption 7 (Nuisance learners). The cross-fitted learners for $\mathbb{E}[T | F]$ and $\mathbb{E}[Y | F]$ have L^2 errors r_T, r_Y ($r_h = r_Y + |\theta|r_T$), and the embargo exceeds the mixing time.

Pooled law for $K \geq 2$ regimes. For homogeneous effects and $w \equiv 1$, Theorem 4 extends to $K \geq 2$: posterior-feature DML satisfies $\hat{\theta} - \theta = a^\top \Sigma_N b / (a^\top \Sigma_N a + \sigma^2) + O_P(N^{-1/2} + r_T r_h + \varepsilon_\tau)$, where D_t is the $(K-1)$ -vector of regime indicators, $\tilde{D}_t = D_t - \mathbb{E}[D_t | F_t]$, $\Sigma_N = N^{-1} \sum_t \tilde{D}_t \tilde{D}_t^\top$, and $T = m(X) + a^\top D + V$, $Y = \theta T + g(X) + b^\top D + U$. Calibration is not needed for this form, because Σ_N is defined relative to the conditional mean given F (Remark: the relevant Σ_N is that of the learner's projection when the learner class is restricted). *Proof. Step 0 (exact decomposition).* By Assumption 5, $\mathbb{E}[T_t | F_t] = m(X_t) + a^\top c_t$ and $\mathbb{E}[Y_t | F_t] = \theta \mathbb{E}[T_t | F_t] + g(X_t) + b^\top c_t$ with $c_t = \mathbb{E}[D_t | F_t]$. With nuisance errors δ_t^T, δ_t^Y and $\delta^h = \delta^Y - \theta \delta^T$, $\tilde{T}_t = a^\top \tilde{D}_t + V_t - \delta_t^T$ and $\tilde{Y}_t - \theta \tilde{T}_t = b^\top \tilde{D}_t + U_t - \delta_t^h$, and $\hat{\theta} - \theta = \sum_t \tilde{T}_t (\tilde{Y}_t - \theta \tilde{T}_t) / \sum_t \tilde{T}_t^2$ exactly. *Step 1.* $N^{-1} \sum (a^\top \tilde{D}_t)(b^\top \tilde{D}_t) = a^\top \Sigma_N b$, $N^{-1} \sum (a^\top \tilde{D}_t)^2 = a^\top \Sigma_N a$, and $N^{-1} \sum V_t^2 = \sigma^2 + O_P(N^{-1/2})$. *Step 2.* Conditionally on $\mathcal{G} = \sigma(S, Z, X)$, $\sum_t V_t b^\top \tilde{D}_t$, $\sum_t a^\top \tilde{D}_t U_t$ and $\sum_t V_t U_t$ have mean zero and variance $O(N)$, hence are $O_P(N^{1/2})$. *Step 3.* Within a fold, $\delta_t = \delta(F_t)$ is fixed given the training folds and $\mathbb{E}[\tilde{D}_t | F_t] = 0$. So $\sum \delta_t \tilde{D}_t$, $\sum \delta_t V_t$ and $\sum \delta_t U_t$ have conditional mean $O(\varepsilon_\tau)$ and variance $\leq CN \|\delta\|_{L^2}^2$ by Davydov's inequality; and $|\sum \delta_t^T \delta_t^h| \leq N r_T r_h$. *Step 4.* Write the ratio as $(A + e_1)/(B + e_2)$ with $A = a^\top \Sigma_N b$, $B = a^\top \Sigma_N a + \sigma^2 \geq \sigma^2$ and $e_i = O_P(N^{-1/2} + r_T r_h + \varepsilon_\tau)$. Then $(A + e_1)/(B + e_2) - A/B = (e_1 - (A/B)e_2)/(B + e_2)$, with $|A/B| \leq \sqrt{b^\top \Sigma_N b}/(2\sigma)$ bounded. $\square$

Hump, sup bound and monotonicity ($K = 2$). (i) $a^2 v + \sigma^2 \geq 2|a|\sqrt{v}\sigma$, with equality at $|a| = \sigma/\sqrt{v}$, so $|B| \leq |\Delta_g|\sqrt{v}/(2\sigma)$. (ii) With $\lambda = \text{Var}(\tilde{T}) = a^2 v + \sigma^2$, $|B| = |\Delta_g|\sqrt{v}\sqrt{\lambda - \sigma^2}/\lambda$, which increases in λ on $[\sigma^2, 2\sigma^2]$ and decreases afterwards, while λ increases monotonically in $|a|$. Conditioning therefore cannot be a monotone certificate. (iii) $\partial_v[v/(a^2 v + \sigma^2)] = \sigma^2/(a^2 v + \sigma^2)^2 > 0$. For $\mathcal{F} \subseteq \mathcal{F}'$, the law of total variance gives $\mathbb{E}\text{Cov}(D | \mathcal{F}') \preceq \mathbb{E}\text{Cov}(D | \mathcal{F})$; a hard label is a measurable function of $\hat{\gamma}$, so the soft feature is never worse for $K = 2$. $\square$

Observable plug-in. Let $CE = \mathbb{E}\|\hat{\gamma}_t - \mathbb{E}[D_t | F_t]\|^2$. With $c = \mathbb{E}[D | F]$, $\gamma(1 - \gamma) - c(1 - c) = (\gamma - c)(1 - 2c) - (\gamma - c)^2$, so $|\mathbb{E}\hat{v} - v| \leq \sqrt{CE} + CE$. If m is linear and the posterior calibrated, the OLS coefficient $\hat{a}$ of $\hat{\gamma}$ in the regression of T on $(1, X, \hat{\gamma})$ is consistent for a , and $\sigma^2 = \text{Var}(\tilde{T}) - a^\top \Sigma a$. $\square$

Heterogeneous effects ($K = 2$). With $\theta(S) \in \{\theta_0, \theta_1\}$, $\bar{\theta} = \mathbb{E}\theta(S)$, $T_t = \bar{m} + m_0(X_t) + aD_t + V_t$, $\mathbb{E}m_0 = 0$ and $\mathbb{E}[m_0(X)\text{Var}(D | F)] = 0$, $\hat{\theta} \rightarrow_p \bar{\theta} + av[b + a(\theta_1 - \bar{\theta}) + \bar{m}(\theta_1 - \theta_0)]/(a^2 v + \sigma^2)$. Expanding $\mathbb{E}[\tilde{T}\theta(S)T]$ with $\tilde{T} = a\tilde{D} + V$ gives $a\bar{m}(\theta_1 - \theta_0)v + a^2\theta_1 v + \sigma^2\bar{\theta}$; subtract $\bar{\theta}(a^2 v + \sigma^2)$. This is the form used for Table 2's semi-synthetic prediction and for the frontier DGP. $\square$

Learned HMM parameters. *Setting (E).* The chain has finitely many states with $A_{ij}^* \geq \varepsilon^* > 0$; the emissions are Gaussian with parameters in a compact set and variances bounded below; and ϑ^* is interior, identifiable

up to labels, with nonsingular Fisher information. Let γ_t^* be the smoothing posterior at ϑ^* and $\hat{\gamma}_t$ at $\hat{\vartheta}$. (H1) $\|\hat{\vartheta} - \vartheta^*\| = O_P(N^{-1/2})$ (and $\mathbb{E}\|\sqrt{N}(\hat{\vartheta} - \vartheta^*)\|^4 \leq C$ for statements in expectation). This is a hypothesis on the output of EM, which we check by multistart.

Lemma 3 (Derivative of the smoother). *Under (E) there are $r > 0$, $C < \infty$ with $D_t := \sup_{\|\vartheta - \vartheta^*\| \leq r} \|\partial_{\vartheta} \gamma_{t,\vartheta}\| \leq C \sum_s \kappa^{|t-s|} (1 + \|Z_s\|^2)$, $\kappa = 1 - \varepsilon^*/2$, and $|\gamma_t^* - \gamma_t^{*,(m)}| \leq \kappa^m$ for the window- m posterior; hence $\sup_{t,N} \mathbb{E} D_t^p < \infty$ and $N^{-1} \sum_t D_t^2 = O_P(1)$.*

Proof. Given Z , the hidden chain has kernels $K_s(i, j) \geq (\varepsilon^*/2)w_j / \sum_l w_l$, a Doeblin minorisation independent of the emissions, so its Dobrushin coefficient is at most κ and conditional covariances decay as $\kappa^{|t-s|}$. Fisher’s identity writes $\partial_{\vartheta} \gamma_t(k) = \sum_s \text{Cov}(\mathbb{I}\{S_t = k\}, h_s \mid Z)$ with $\text{osc}(h_s) \leq C(1 + \|Z_s\|^2)$; summing gives the bound, and the same contraction gives forgetting. $\square$

Proposition 3 (Rate for learned parameters). *Assume (E) and (H1), and let $\delta_X = N^{-1} \sum_t \mathbb{E}(\mathbb{E}[D_t \mid Z, X] - \gamma_t^*)^2$ be the regime information carried by the controls beyond the proxies. Then $N^{-1} \sum_t \|\hat{\gamma}_t - \gamma_t^*\|^2 = O_P(N^{-1})$, $\|\mathbb{E}\hat{v} - v^*\| = O(N^{-1/2})$, $CE \leq (\sqrt{\delta_X} + O(N^{-1/2}))^2$ and $|v_N - v^*| \leq 2\sqrt{\delta_X} + O(N^{-1/2})$. The plug-in is therefore $N^{-1/2}$ -accurate when $\delta_X = 0$, and its guarantee degrades by $\sqrt{\delta_X}$ otherwise.*

Proof. On $\{\|\hat{\vartheta} - \vartheta^*\| \leq r\}$ the mean-value theorem and Lemma 3 give $\|\hat{\gamma}_t - \gamma_t^*\| \leq D_t \|\hat{\vartheta} - \vartheta^*\|$, and averaging gives the first claim. Since $|x(1-x) - y(1-y)| \leq |x-y|$ the second follows. With P the L^2 projection onto $\sigma(X, \hat{\gamma})$, $CE = \|P(\hat{\gamma} - \gamma^X)\|^2 \leq \|\hat{\gamma} - \gamma^X\|^2$, and $|v_N - v^*| \leq \|c - \gamma^*\| \leq \|\hat{\gamma} - \gamma^X\| + \sqrt{\delta_X}$. $\square$ *Empirical checks* (supplement, [reports/bias_law/x21-x30](#)). When the emission model is correct, $\mathbb{E}|\hat{\gamma} - \gamma^*|$ decays with log-log slope -0.49 . Under misspecified (correlated t_5) emissions the plug-in $\hat{v}$ equals the pseudo-true value but understates the realized residual variance by a factor of two, so it is a lower bound. Refitting the HMM on training folds only leaves the bias unchanged (correlation 0.999 between fold-wise and global estimates). Mumbai’s likelihood is multimodal: five restarts span 1,182 log-likelihood units, so (H1) fails there, and the best restart changes its robustness value from 11.9 to 7.1 SD without changing any conclusion. On the real series $\delta_X > 0$, because the controls contain the proxy variables.

A priori residual variance. For a continuous AR(1) latent state with coefficient ϕ observed with information rate $\iota = h^\top R^{-1}h$, the Kalman-smoothed residual variance is $v = (1 - \phi^2) / \sqrt{(1 + \phi^2 + \iota(1 - \phi^2))^2 - 4\phi^2}$ (Wiener–Kolmogorov). The worst-case bias therefore decays only as $\iota^{-1/4}$ while $\iota \ll (1 - \phi^2)^{-1}$: halving it needs about $16\times$ the sensor information. For a persistent two-state chain with Mahalanobis separation d and switch rate r , residual variance is paid at change points, $v \approx c r e^{-d^2/8} / d$ with $c \rightarrow \sqrt{\pi/2}$ for an isolated switch. Discrete regimes with informative proxies are therefore far less dangerous than a continuous latent state with the same sensor information. Statements, proofs, numerical checks and fitted constants are in the supplementary archive ([archive/manuscript_lineages/v2.bias_law_main.tex](#), [reports/bias_law/x6](#), [x15-x20](#), [x28](#), [x31](#)).

G.1 Experiments for the bias law

All use `src/bias_law/`: a persistent chain (stay probability 0.9), 2-d Gaussian proxies, $N = 3000$, $\Delta_g = 3$, an EM-fitted HMM, five purged folds with $\tau = 24$, and 12 replications per cell. Across 72 two-regime cells (smoother and filter; linear and boosted nuisances; $\Delta_Z \in \{0.5, 0.8, 1.2\}$; eight values of Δ_T from 0.25 to 32), the law with the observable plug-in $\hat{v}$ has correlation 0.996 and RMSE 0.012 with the measured bias, and median relative error 5.2% on cells with $|\text{bias}| > 0.05$ (3.5% with the realized v_N ; Table 6). Subtracting the plug-in bias with the true b reduces mean absolute error from 0.126 to 0.021. Table 7 shows the pitfall behind “conditioning sweeps”: if nuisances omit the regime, $\lambda_{\min}$ grows from 0.54 to 14.4 while error is hump-shaped (peak 14.0); with regime-aware nuisances $\lambda_{\min}$ is flat at 0.51 and the error stays in $[0.06, 0.10]$. Table 8 shows the law explaining the frontier DGP of Table 2. Table 9 shows that temperature scaling changes $\hat{v}$ by a factor of 14 while the realized bias is constant.

H INFERENCE

Scope. Theorem 5 covers HMM parameters fitted on a calibration window separated from the evaluation sample (causal filtering or a held-out fit). Our real-data analyses use full-sample smoothing, which the theorem does not cover. In the bias-law designs, refitting the HMM fold-wise instead of globally leaves estimates unchanged (correlation 0.999; Appendix G), which is evidence, not a proof, that the gap is immaterial there. The proof

Table 6: Agreement between measured bias and the law (bias-law experiments E1–E2).

Setting	cells	corr($\hat{\theta} - \theta$, law($\hat{v}$))	RMSE	med. rel. err.	corr (law(v_N))
$K=2$, filter, lin	24	0.997	0.016	7.3%	1.000
$K=2$, smooth, gbm	24	0.998	0.011	4.3%	0.995
$K=2$, smooth, lin	24	0.998	0.009	5.1%	0.999
$K=3$, smooth, lin	8	0.999	0.010	9.8%	0.998

 Table 7: Conditioning sweep on the frontier DGP ($\Delta_Z = 1.5$): the role of regime-aware nuisances.

Δ_T	nuisances on X only		nuisances on (X, S)	
	$\lambda_{\min}(\hat{J})$	$\ \hat{\theta} - \theta\ _2$	$\lambda_{\min}(\hat{J})$	$\ \hat{\theta} - \theta\ _2$
0.4	0.54	5.92	0.51	0.10
0.8	0.61	10.34	0.51	0.08
1.5	0.86	14.05	0.51	0.07
3.0	1.88	13.38	0.51	0.07
6.0	5.61	8.76	0.51	0.07
10.0	14.38	5.73	0.51	0.06

below is written for posterior-weighted nuisances, where condition (G) controls the non-orthogonal Gram part; for posterior-adjusted nuisances Lemma 1 makes the score orthogonal in both nuisances, (G) is not needed, and Step 2 holds with remainder $\mathcal{O}_P(r_N^2)$.

Assumption 8 (HMM Regularity, Gram-Nuisance Rate, Moments). (H) The emission model is identifiable up to label permutation; $\hat{\mathbf{A}}$ is the MLE fitted on a calibration window of length N_{cal} separated by the embargo, with $N/N_{\text{cal}} \rightarrow \kappa \in [0, \infty)$ ($\kappa = 0$ for oracle filter), and $\gamma_{tk}(\mathbf{A})$ is continuously differentiable with bounded derivative. Then $\sqrt{N_{\text{cal}}}(\hat{\mathbf{A}} - \mathbf{A}^*) \Rightarrow \mathcal{N}(\mathbf{0}, \mathcal{I}_{\mathbf{A}}^{-1})$ (Leroux, 1992; Bickel et al., 1998; Douc et al., 2004). (G) $\sqrt{\varepsilon_\gamma} \max_k \|\hat{m}_k - m_k^\gamma\|_{L_2} = o_P(N^{-1/2})$; true for hard assignments and under Assumption 4 when $\varepsilon_\gamma = \mathcal{O}(N^{-1/2})$. (M) $\mathbb{E}\|\Psi_t\|^{2+\delta} < \infty$ for some $\delta > 0$, and $\lambda_{\min}(\mathbf{J}) + \lambda \geq c_\lambda > 0$ is fixed.

Proof of Theorem 5. Step 1 (exact algebra, centering at θ_λ). Because $(\hat{\mathbf{J}} + \lambda \mathbf{I})\hat{\theta}_\lambda = \hat{\mathbf{S}}$ and $(\mathbf{J} + \lambda \mathbf{I})\theta_\lambda = \mathbf{S}$:

$$\hat{\theta}_\lambda - \theta_\lambda = (\hat{\mathbf{J}} + \lambda \mathbf{I})^{-1} \left[(\hat{\mathbf{S}} - \hat{\mathbf{J}}\theta_\lambda) - (\mathbf{S} - \mathbf{J}\theta_\lambda) \right] = (\hat{\mathbf{J}} + \lambda \mathbf{I})^{-1} \frac{1}{N} \sum_{t=1}^N \tilde{\Psi}_t, \quad (23)$$

with $\tilde{\Psi}_t = \Psi_t(\theta_\lambda) - \mathbb{E}\Psi_t(\theta_\lambda)$ strictly mean zero. (At θ^* the summands have non-vanishing mean $\mathbf{b}_\gamma$, so the CLT cannot be centered there without rate conditions on ε_γ .)

Step 2 (nuisance errors). Expanding in $\hat{\eta} - \eta^\gamma$: by Lemma 2(i)-(ii) the μ -direction and cross-moment have zero first-order term; the Gram part contributes $\mathcal{O}(\sqrt{\varepsilon_\gamma} \|\hat{m} - m^\gamma\|) = o_P(N^{-1/2})$ by (G); second-order remainder is $\mathcal{O}_P(r_N^2) = o_P(N^{-1/2})$ by Assumption 4.

Step 3 (cross-fitting under dependence). By Bradley’s coupling lemma (Bradley, 2005), nuisance fits on blocks separated by τ^* from test blocks match an independent copy except on an event of probability $\leq N\alpha(\tau^*) \leq MN^{1-c_\alpha C} \rightarrow 0$ when $C > 1/c_\alpha$.

Step 4 (CLT). $\tilde{\Psi}_t$ is strictly stationary, mean zero, α -mixing with geometric rate and $(2+\delta)$ moments, so $N^{-1/2} \sum_t \tilde{\Psi}_t \Rightarrow \mathcal{N}(\mathbf{0}, \mathbf{\Omega}_\lambda)$ (Ibragimov, 1962).

Step 5 (HMM parameters). With estimated $\hat{\mathbf{A}}$, $N^{-1/2} \sum_t \tilde{\Psi}_t(\hat{\mathbf{A}}) = N^{-1/2} \sum_t \tilde{\Psi}_t(\mathbf{A}^*) + \mathbf{A}_\Lambda \sqrt{N}(\hat{\mathbf{A}} - \mathbf{A}^*) + o_P(1)$. By (H), $\sqrt{N}(\hat{\mathbf{A}} - \mathbf{A}^*) \Rightarrow \mathcal{N}(\mathbf{0}, \kappa \mathcal{I}_{\mathbf{A}}^{-1})$, asymptotically independent of the evaluation sum across the embargo buffer, so the asymptotic covariances add: $\mathbf{\Omega}_\lambda^{\text{tot}} = \mathbf{\Omega}_\lambda + \kappa \mathbf{A}_\Lambda \mathcal{I}_{\mathbf{A}}^{-1} \mathbf{A}_\Lambda^\top$.

Step 6 (Slutsky). $\hat{\mathbf{J}} \rightarrow_p \mathbf{J}$ and $\lambda_{\min}(\mathbf{J}) + \lambda \geq c_\lambda$ yield $(\hat{\mathbf{J}} + \lambda \mathbf{I})^{-1} \rightarrow_p (\mathbf{J} + \lambda \mathbf{I})^{-1}$. (b) is the identity $\theta_\lambda - \theta^* = \text{Bias}$; (c) follows by Slutsky because $\sqrt{N}\|\text{Bias}\| \leq \sqrt{N}(C\varepsilon_\gamma + \lambda M_\theta)/(\lambda_{\min} + \lambda)$. ■

Operational Modes of Forward Filtering. In offline retrospective smoothing, $\gamma_{tk}^{\text{smooth}} = P(S_t = k \mid Z_{1:N})$. In real-time decision making, three distinct forward modes exist: (1) *Oracle Filter*: $P(S_t = k \mid Z_{1:t}, \mathbf{A}^*)$ using the

Table 8: Frontier DGP (30 replications, $N = 1200$): excess bias of soft conditioning over the oracle state, and the law.

Δ_Z	DML on X	soft $\hat{\gamma}$	oracle S	excess bias	law(v_N)	law($\hat{v}$)
0.2	10.02	9.773	1.535	8.238	8.236	7.418
0.5	10.02	7.776	1.535	6.241	6.267	5.655
1.0	10.02	3.796	1.535	2.260	2.299	1.891
1.5	10.02	1.923	1.535	0.388	0.402	0.320
2.0	10.02	1.559	1.535	0.024	0.024	0.016
4.0	10.02	1.535	1.535	0.000	0.000	0.000

 Table 9: Temperature scaling of the posterior ($K = 2$, boosted nuisances, 20 replications, $\Delta_Z = 0.8$, $\Delta_T = 2$).

τ	bias	law($\hat{v}$)	law(v_N)	$\hat{v}$	v_N	$\sqrt{CE} + CE$
0.30	0.128	0.033	0.160	0.006	0.030	0.071
0.50	0.127	0.056	0.160	0.010	0.030	0.053
0.75	0.127	0.088	0.160	0.016	0.030	0.035
1.00	0.127	0.123	0.160	0.022	0.030	0.025
1.50	0.127	0.201	0.160	0.039	0.030	0.038
2.50	0.127	0.355	0.160	0.077	0.030	0.098

true HMM parameters; (2) *Training-Fitted Forward Filter*: $P(S_t = k \mid Z_{1:t}, \hat{\Lambda})$ where parameters $\hat{\Lambda}$ are fitted on a historical calibration window strictly preceding the evaluation window ($\mathcal{T}_{\text{cal}} < \mathcal{T}_{\text{eval}}$) and state inference updates strictly forward on past-to-present observations $Z_{1:t}$; and (3) *Expanding-Window Online Filter*: where both parameters $\hat{\Lambda}_t$ and state updates proceed sequentially. Our implementations evaluate Mode 1 and Mode 2, where Mode 2 enforces causal forward state filtering without future observation leakage (distinguishing forward state updating from online sequential parameter re-estimation).

H.1 Observed-Data Orthogonal Score under Estimated Latent Dynamics

In semiparametric causal inference (Chernozhukov et al., 2018; Bickel et al., 1993), when nuisance parameters consist of both non-parametric outcome/propensity regressions $\eta = (m, \mu)$ and finite-dimensional parameters Λ governing latent state transitions and emissions, one must characterize the sensitivity of the causal estimator to the first-order estimation error in $\hat{\Lambda}$.

Let the state-weighted regime score vector be:

$$\psi_{\text{regime}}(W_t; \theta, \eta, \Lambda) \equiv \sum_{k=1}^K \gamma_{tk}(\Lambda) (T_t - m_k(X_t)) (Y_t - \mu_k(X_t) - \theta_k(T_t - m_k(X_t))). \quad (24)$$

In a hidden Markov model satisfying standard regularity and mixing conditions (Bickel et al., 1993), the latent parameters Λ are estimated via sequence-level maximum likelihood with sequence score $\nabla_{\Lambda} \log p(Z_{1:N}; \Lambda) = \sum_{t=1}^N \phi_{\Lambda,t} + o_P(\sqrt{N})$, where $\phi_{\Lambda,t} \equiv \nabla_{\Lambda} \log p(Z_t \mid Z_{1:t-1}; \Lambda^*)$ denotes the sequence-level prediction-error score increment satisfying $\mathbb{E}[\phi_{\Lambda,t} \mid Z_{1:t-1}] = \mathbf{0}$ and $\frac{1}{N} \sum_{t=1}^N \mathbb{E}[\phi_{\Lambda,t} \phi_{\Lambda,t}^\top] \rightarrow \mathcal{I}_{\Lambda}$. Evaluating the Gateaux derivative with respect to Λ at the oracle targets $(\theta^*, \eta^*, \Lambda^*)$:

$$\mathbf{A}_{\Lambda} \equiv \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{t=1}^N \sum_{k=1}^K \mathbb{E} \left[\frac{\partial \gamma_{tk}(\Lambda^*)}{\partial \Lambda^\top} (T_t - m_k^*(X_t)) (Y_t - \mu_k^*(X_t) - \theta_k^*(T_t - m_k^*(X_t))) \right]. \quad (25)$$

Importantly, $\mathbf{A}_{\Lambda}$ does not vanish pointwise across regimes. While Assumption 3 guarantees that the structural innovation has zero mean conditional on the true state, $\mathbb{E}[Y_t - \mu_k^*(X_t) - \theta_k^*(T_t - m_k^*(X_t)) \mid X_t, T_t, S_t = k] = 0$, conditioning on observed history $Z_{1:N}$ alone yields:

$$\mathbb{E}[Y_t - \mu_k^*(X_t) - \theta_k^*(T_t - m_k^*(X_t)) \mid X_t, T_t, Z_{1:N}] = \sum_{j \neq k} \gamma_{tj}(\Lambda^*) [(\mu_j^*(X_t) - \mu_k^*(X_t)) + (\theta_j^* - \theta_k^*)(T_t - m_j^*(X_t))]. \quad (26)$$

Table 10: When can b be learned? Posterior-DML versus a correctly specified joint Markov-switching MLE and proximal 2SLS with a negative-control outcome (bias, RMSE in parentheses).

Scenario	Δ_T	Posterior-DML	Joint MLE	Proximal 2SLS
HMM correctly specified	1	+0.20 (0.20)	+0.00 (0.02)	-0.01 (0.02)
HMM correctly specified	4	+0.40 (0.40)	+0.00 (0.02)	+0.02 (0.06)
nonlinear $f(X)$ (GBM nuisances)	1	+0.16 (0.17)	-0.01 (0.01)	-0.00 (0.04)
nonlinear $f(X)$ (GBM nuisances)	4	+0.37 (0.37)	-0.00 (0.01)	-0.03 (0.06)
heavy-tailed (t_3) proxy noise	1	+0.15 (0.16)	+0.01 (0.02)	+0.00 (0.04)
heavy-tailed (t_3) proxy noise	4	+0.32 (0.32)	-0.00 (0.01)	-0.04 (0.05)
true $K=3$, fitted $K=2$	1	+0.46 (0.47)	-0.00 (0.02)	-0.01 (0.05)
true $K=3$, fitted $K=2$	4	+0.56 (0.57)	+0.00 (0.01)	+0.06 (0.16)
heterogeneous $\theta(S)$ (ATE target)	1	+0.26 (0.26)	-0.01 (0.03)	-0.02 (0.06)
heterogeneous $\theta(S)$ (ATE target)	4	+0.66 (0.66)	-0.00 (0.02)	-0.01 (0.08)
negative control leaks ($W \ni 0.3T$)	1	+0.21 (0.21)	+0.01 (0.03)	-0.45 (0.45)
negative control leaks ($W \ni 0.3T$)	4	+0.39 (0.39)	-0.00 (0.01)	-0.46 (0.47)
continuous AR(1) latent, SNR 0.5	1	+0.83 (0.84)	+0.79 (0.80)	-0.02 (0.04)
continuous AR(1) latent, SNR 1	1	+0.77 (0.78)	+0.79 (0.79)	-0.02 (0.04)
continuous AR(1) latent, SNR 0.5	2	+0.90 (0.90)	+0.89 (0.89)	-0.03 (0.08)
continuous AR(1) latent, SNR 1	2	+0.87 (0.88)	+0.88 (0.89)	-0.02 (0.06)

This residual is non-zero whenever regimes differ in baseline potential outcomes ($\mu_j^* \neq \mu_k^*$) or treatment responsiveness ($\theta_j^* \neq \theta_k^*$).

Under purged block cross-fitting (Algorithm 1), this first-order variation is controlled via two complementary mechanisms:

1. **Decoupling via Purged Embargo Buffer:** The parameters $\hat{\Lambda}$ are estimated on historical calibration blocks or out-of-fold data separated by an embargo buffer $\tau^* \geq C \log N$. By the α -mixing decay of the underlying process (Assumption 4), the covariance between evaluation fold innovations and the parameter estimation error $\hat{\Lambda} - \Lambda^*$ decays as $O(N^{-C\alpha})$.
2. **Stacked Joint HAC Variance Representation:** When $\hat{\Lambda}$ is estimated at parametric rate $\sqrt{N}(\hat{\Lambda} - \Lambda^*) = \mathcal{I}_\Lambda^{-1} \frac{1}{\sqrt{N}} \sum_{t=1}^N \phi_{\Lambda,t} + o_P(1)$, first-order parameter variation is absorbed into the sequence-level influence representation $\phi_t = \phi_{\theta,t} + \mathbf{A}_\Lambda \mathcal{I}_\Lambda^{-1} \phi_{\Lambda,t}$, yielding the stacked M-estimation limiting distribution:

$$\sqrt{N} \begin{pmatrix} \hat{\theta}_\lambda - \theta^* - \text{Bias}(\lambda) \\ \hat{\Lambda} - \Lambda^* \end{pmatrix} \xrightarrow{d} \mathcal{N} \left(\mathbf{0}, \begin{bmatrix} \Sigma_\lambda & \Sigma_{\theta\Lambda} \\ \Sigma_{\theta\Lambda}^\top & \mathcal{I}_\Lambda^{-1} \end{bmatrix} \right), \quad (27)$$

where $\Sigma_{\theta\Lambda} \equiv \text{LRCov}(\phi_{\theta,t}, \phi_{\Lambda,t})$, and the long-run covariance Σ_λ and cross-term $\Sigma_{\theta\Lambda}$ are consistently captured by the joint Newey-West HAC sandwich estimator. In well-separated regimes ($\Delta_Z \rightarrow \infty$), posterior uncertainty vanishes ($\gamma_{tk} \rightarrow \mathbb{I}(S_t = k)$) and $\partial \gamma_{tk} / \partial \Lambda \rightarrow \mathbf{0}$ exponentially fast, rendering $\|\mathbf{A}_\Lambda\| \leq C e^{-c\Delta_Z^2}$ and recovering asymptotic orthogonality.

H.2 Regime occupation and the population ATE

We distinguish three distinct regime weighting vectors:

1. **True Stationary Distribution π^* :** The invariant measure satisfying $\pi^{*\top} \Pi^* = \pi^{*\top}$.
2. **Estimated HMM Stationary Distribution $\hat{\pi}(\Lambda)$:** The stationary vector derived from estimated HMM matrix $\hat{\Pi}$.
3. **Sample Posterior Occupation Distribution $\bar{\gamma}_N$:** The empirical average $\bar{\gamma}_N \equiv \frac{1}{N} \sum_{t=1}^N \gamma_t$.

The sample Average Treatment Effect estimator is $\hat{\theta}_{\text{SATE}} = \sum_{k=1}^K \bar{\gamma}_{N,k} \hat{\theta}_{\lambda,k} = \bar{\gamma}_N^\top \hat{\theta}_\lambda$, with conditional variance $\sigma_{\text{SATE}}^2 = \bar{\gamma}_N^\top \Sigma_\lambda \bar{\gamma}_N$. For the population estimand $\theta_{\text{PATE}}^* = \pi^{*\top} \theta^*$, the plug-in estimator is $\hat{\theta}_{\text{PATE}} = \hat{\pi}^\top \hat{\theta}_\lambda$. Expanding $\hat{\theta}_{\text{PATE}}$ around (π^*, θ^*) via the multivariate delta method: $\sqrt{N}(\hat{\theta}_{\text{PATE}} - \theta_{\text{PATE}}^*) = \pi^{*\top} \sqrt{N}(\hat{\theta}_\lambda - \theta^*) + \theta^{*\top} \sqrt{N}(\hat{\pi} - \pi^*) + o_P(1)$, yielding asymptotic variance:

$$\sigma_{\text{PATE}}^2 = \lim_{N \rightarrow \infty} \text{Var} \left(\sqrt{N}(\hat{\theta}_{\text{PATE}} - \theta_{\text{PATE}}^*) \right) = \pi^{*\top} \Sigma_\lambda \pi^* + \theta^{*\top} \Sigma_\pi \theta^* + 2\pi^{*\top} \Sigma_{\theta\pi} \theta^*, \quad (28)$$

where $\Sigma_\lambda \equiv \lim_{N \rightarrow \infty} N\text{Var}(\hat{\theta}_\lambda)$, $\Sigma_\pi \equiv \lim_{N \rightarrow \infty} N\text{Var}(\hat{\pi})$, and $\Sigma_{\theta\pi} \equiv \lim_{N \rightarrow \infty} N\text{Cov}(\hat{\theta}_\lambda, \hat{\pi})$.

Cross-Covariance Analysis and HMM Influence Orthogonality: When $\hat{\theta}_\lambda$ and $\hat{\pi}$ share estimated parameters $\hat{\Lambda}$, let $\phi_\theta(W_t) \equiv (\mathbf{J} + \lambda \mathbf{I})^{-1} \Psi_t$ and $\phi_{\Lambda,t} \equiv \nabla_{\Lambda} \log p(Z_t \mid Z_{1:t-1}; \Lambda^*)$. With $\sqrt{N}(\hat{\pi} - \pi^*) = \nabla_{\Lambda} \pi^* \mathcal{I}_{\Lambda}^{-1} \frac{1}{\sqrt{N}} \sum_{t=1}^N \phi_{\Lambda,t} + o_P(1)$, the cross-covariance evaluates to:

$$\Sigma_{\theta\pi} = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{t=1}^N \sum_{s=1}^N \mathbb{E} \left[\phi_\theta(W_t) \phi_{\Lambda,t}^\top \right] \mathcal{I}_{\Lambda}^{-1} (\nabla_{\Lambda} \pi^*)^\top = \text{LRCov}(\phi_\theta, \phi_{\Lambda}) \mathcal{I}_{\Lambda}^{-1} (\nabla_{\Lambda} \pi^*)^\top. \quad (29)$$

A sufficient condition for $\Sigma_{\theta\pi} = \mathbf{0}$ is long-run cross-orthogonality $\text{LRCov}(\phi_\theta, \phi_{\Lambda}) = \mathbf{0}$. In practice, σ_{PATE}^2 is consistently estimated via a joint Newey-West HAC sandwich over the stacked $(2K)$ -dimensional influence sequence $(\phi_\theta(W_t)^\top, \phi_\pi(Z_t)^\top)^\top$, where $\phi_\pi(Z_t) \equiv \nabla_{\Lambda} \pi^* \mathcal{I}_{\Lambda}^{-1} \phi_{\Lambda,t}$.

Stationary Distribution Variance vs. Sample Regime Occupancy: The parametric variance of $\hat{\pi}(\hat{\Lambda})$ is $\Sigma_\pi = \nabla_{\Lambda} \pi^* \mathcal{I}_{\Lambda}^{-1} (\nabla_{\Lambda} \pi^*)^\top$. For the sample occupancy process $\bar{\mathbf{S}}_N \equiv \frac{1}{N} \sum_{t=1}^N \mathbf{H}_t$, asymptotic covariance is governed by fundamental matrix $\mathbf{Z} \equiv (\mathbf{I} - \mathbf{\Pi}^* + \mathbf{1}\pi^{*\top})^{-1}$ (Kemeny and Snell, 1976): $\Sigma_{\text{occ}} = \mathbf{D}_\pi \mathbf{Z} + \mathbf{Z}^\top \mathbf{D}_\pi - \mathbf{D}_\pi - \pi^* \pi^{*\top}$, with $\mathbf{D}_\pi = \text{diag}(\pi^*)$. For a two-state Markov chain ($K = 2$) with persistence $\rho = \lambda_2(\mathbf{\Pi}^*)$:

$$\text{Var}(\bar{S}_N) = \frac{\pi_0^* \pi_1^*}{N} \left(\frac{1 + \rho}{1 - \rho} \right) + o(N^{-1}). \quad (30)$$

When $\Delta\theta^* \neq 0$, Markov persistence inflates the sampling variance of $\hat{\theta}_{\text{SATE}}$ by $\frac{1+\rho}{1-\rho} \approx 15.7\times$ under persistent stagnation ($\rho \approx 0.88$) relative to independent transitions. ■

I OPERATIONAL SURROGATE, SCREENING AND REGULARIZATION

Assumption 9 (Two-sided spectral comparability). There exist $0 < \underline{c}_J \leq \bar{c}_J < \infty$ with $\underline{c}_J \leq \lambda_{\min}(\mathbf{J}_t)/\lambda_{\min}(\mathbf{J}) \leq \bar{c}_J$ for rolling Gram matrices $\mathbf{J}_t$, $t \geq w$.

Proposition 4 (Entropy surrogate). For calibrated forward posteriors in binary regimes, $\mathbb{E}[\|\gamma_t - \mathbf{H}_t\|_1 \mid \mathcal{F}_t] = 2 \text{Gini}(\gamma_t) \leq C_K H(\gamma_t)$ with $C_K \leq 1/\ln 2$. Hence $\mathcal{D}_{\text{causal}} \equiv \varepsilon_\gamma / \lambda_{\min}(\mathbf{J}) \leq C_K \mathbb{E}[H(\gamma_t)] / \lambda_{\min}(\mathbf{J})$, and under Assumption 9 $\mathcal{D}_{\text{causal}} \leq C_K \bar{c}_J \mathbb{E}[H(\gamma_t)] / \lambda_{\min}(\mathbf{J}_t)$.

The surrogate is an upper bound on $\mathcal{D}_{\text{causal}}$, which is defined with the *generated-state* $\mathbf{J}$. By Section 4 that object can be inflated relative to the oracle geometry, so the surrogate inherits phantom resolution. We keep it as a monitoring device, not a certificate. Let $\gamma_t \equiv P(S_t \in \cdot \mid \mathcal{F}_t) \in \Delta^K$ be the Bayesian posterior state distribution with respect to filtration $\mathcal{F}_t \equiv \sigma(Z_{1:t})$. Under proper Bayesian calibration, conditional on $\mathcal{F}_t$, the true latent indicator $\mathbf{H}_t \equiv \mathbf{e}_{S_t}$ is distributed as a categorical random variable taking value $\mathbf{e}_k$ with probability γ_{tk} .

Step 1: Expected L_1 Error as Gini Impurity. Evaluating the conditional expectation of the L_1 error:

$$\begin{aligned} \mathbb{E}[\|\gamma_t - \mathbf{H}_t\|_1 \mid \mathcal{F}_t] &= \sum_{k=1}^K \gamma_{tk} \|\gamma_t - \mathbf{e}_k\|_1 = \sum_{k=1}^K \gamma_{tk} \left((1 - \gamma_{tk}) + \sum_{j \neq k} \gamma_{tj} \right) \\ &= \sum_{k=1}^K \gamma_{tk} \cdot 2(1 - \gamma_{tk}) = 2 \left(1 - \sum_{k=1}^K \gamma_{tk}^2 \right) = 2 \text{Gini}(\gamma_t). \end{aligned} \quad (31)$$

Step 2: Entropy Bound on Gini Impurity. By the standard concave inequality $\ln(x) \leq x - 1$, for any $x > 0$:

$$H(\gamma_t) = - \sum_{k=1}^K \gamma_{tk} \ln \gamma_{tk} = \sum_{k=1}^K \gamma_{tk} \ln \left(\frac{1}{\gamma_{tk}} \right) \geq \sum_{k=1}^K \gamma_{tk} (1 - \gamma_{tk}) = \text{Gini}(\gamma_t). \quad (32)$$

Furthermore, for any binary regime partition ($K = 2$) with $\gamma_{t1} = p$ and $\gamma_{t2} = 1 - p$, $2 \text{Gini}(\gamma_t) = 4p(1 - p)$. Comparing to binary Shannon entropy $H(p) = -p \ln p - (1 - p) \ln(1 - p)$, the ratio $\frac{4p(1-p)}{H(p)}$ achieves its maximum at $p = 1/2$, where $\frac{4(1/4)}{\ln 2} = \frac{1}{\ln 2} \approx 1.4427$. Hence for all $p \in [0, 1]$:

$$2 \text{Gini}(\gamma_t) \leq C_K H(\gamma_t), \quad \text{where } C_K \leq \frac{1}{\ln 2} \approx 1.4427. \quad (33)$$

Step 3: Population and Observable Bridge. Taking the population expectation of Step 1 and applying Step 2:

$$\varepsilon_\gamma \equiv \mathbb{E}[\mathbb{E}[\|\gamma_t - \mathbf{H}_t\|_1 \mid \mathcal{F}_t]] = 2\mathbb{E}[\text{Gini}(\gamma_t)] \leq C_K \mathbb{E}[H(\gamma_t)]. \quad (34)$$

Dividing by the global minimal singular value $\lambda_{\min}(\mathbf{J})$:

$$\mathcal{D}_{\text{causal}} \equiv \frac{\varepsilon_\gamma}{\lambda_{\min}(\mathbf{J})} \leq C_K \mathbb{E}\left[\frac{H(\gamma_t)}{\lambda_{\min}(\mathbf{J})}\right] = C_K \mathbb{E}[\mathcal{D}_{\text{global}}^H(t)]. \quad (35)$$

Furthermore, under Assumption 9 (Two-Sided Spectral Comparability), the rolling Gram matrix satisfies $\lambda_{\min}(\mathbf{J}_t) \leq \bar{c}_J \lambda_{\min}(\mathbf{J})$ almost surely for all evaluation windows $t \geq w$, which implies $\frac{1}{\lambda_{\min}(\mathbf{J})} \leq \frac{\bar{c}_J}{\lambda_{\min}(\mathbf{J}_t)}$. Pointwise substitution yields:

$$\mathcal{D}_{\text{causal}} \leq C_K \bar{c}_J \mathbb{E}\left[\frac{H(\gamma_t)}{\lambda_{\min}(\mathbf{J}_t)}\right] = C_K \bar{c}_J \mathbb{E}[\mathcal{D}_{\text{operational}}(t)], \quad (36)$$

and $\mathbb{E}[\mathcal{D}_{\text{operational}}(t)] \leq \bar{c}_J^{-1} \mathbb{E}[\mathcal{D}_{\text{global}}^H(t)]$. In simulation benchmarks where true states are known and posteriors are evaluated at population parameters Λ^* , the ergodic theorem and central limit theorem for stationary α -mixing chains ensure that $\hat{\varepsilon}_{\gamma,N} \equiv \frac{1}{N} \sum_{t=1}^N \|\gamma_t - \mathbf{H}_t\|_1 = \varepsilon_\gamma + \mathcal{O}_P(N^{-1/2})$ (preserved under $\sqrt{N}$ -consistent estimation of Λ), providing an evaluation-only benchmark surrogate $\hat{\mathcal{D}}_{\text{causal},N} \equiv \hat{\varepsilon}_{\gamma,N} / \lambda_{\min}(\mathbf{J}) = \mathcal{D}_{\text{causal}} + \mathcal{O}_P(N^{-1/2})$. We emphasize that these bounds establish $\mathcal{D}_{\text{operational}}(t)$ as an observable surrogate upper bound for $\mathcal{D}_{\text{causal}}$, not a two-sided equivalence. ■

I.1 A Priori Proxy Diagnostics and Separation Screening

Before fitting OR-DML or downstream prediction heads, practitioners should assess proxy informativeness on auxiliary measurements $Z_t \in \mathbb{R}^q$. Let $\hat{\boldsymbol{\mu}}_{Z,k} \equiv \frac{1}{N_k} \sum_{t:\hat{S}_t=k} Z_t$ and $\hat{\boldsymbol{\Sigma}}_{Z,k}$ denote empirical cluster moments. We define the standardized separation index:

$$\Delta_Z \equiv \min_{j \neq k} \frac{\|\hat{\boldsymbol{\mu}}_{Z,j} - \hat{\boldsymbol{\mu}}_{Z,k}\|_2}{\sqrt{\frac{1}{2}(\text{Tr}(\hat{\boldsymbol{\Sigma}}_{Z,j}) + \text{Tr}(\hat{\boldsymbol{\Sigma}}_{Z,k}))/q}}. \quad (37)$$

When $\Delta_Z < 0.5$ (or silhouette score < 0.15), the overlap between regime emission densities $P(Z_t \mid S_t = j)$ and $P(Z_t \mid S_t = k)$ is severe, driving posterior entropy $\bar{H}(\gamma) \rightarrow \ln K$. By Theorem 2, point identification fails identically in the exact uninformative limit $\Delta_Z = 0$ ($p = 0.5$ or static), while breakdown at small positive separation ($\Delta_Z \leq 1.0$) is governed by finite-sample proxy ill-conditioning under Theorem 3. Deployed pipelines should enforce an automated screening gate: if $\Delta_Z < 0.5$ or mean entropy exceeds $0.80 \ln K$, the system must flag non-identifiability and refuse to emit point causal effects without external structural constraints.

I.2 Spectral regularization

Theorem 6 (Risk bound). *For $\lambda \geq 0$ with $\lambda_{\min}(\mathbf{J}) + \lambda > 0$: $\|\boldsymbol{\theta}_\lambda - \boldsymbol{\theta}^*\|_2 \leq (C\varepsilon_\gamma + \lambda M_\theta) / (\lambda_{\min}(\mathbf{J}) + \lambda)$, and if $N\|\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}_\lambda\|^2$ is uniformly integrable, $\mathbb{E}\|\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}^*\|_2^2 \leq 2B_\lambda^2 + 2\text{Tr}(\boldsymbol{\Omega}_\lambda^{\text{tot}}) / (N(\lambda_{\min} + \lambda)^2) + o(N^{-1})$.*

Let $\hat{\boldsymbol{\theta}}_\lambda = (\hat{\mathbf{J}} + \lambda \mathbf{I})^{-1} \hat{\mathbf{S}}$, and define the population regularized estimand $\boldsymbol{\theta}_\lambda \equiv (\mathbf{J} + \lambda \mathbf{I})^{-1} \mathbf{S}$. We decompose the estimation error around the regularized population estimand $\boldsymbol{\theta}_\lambda$ via the exact algebraic identity:

$$\mathbb{E}\|\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}^*\|_2^2 = \|\boldsymbol{\theta}_\lambda - \boldsymbol{\theta}^*\|_2^2 + \mathbb{E}\|\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}_\lambda\|_2^2 + 2\langle \mathbb{E}[\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}_\lambda], \boldsymbol{\theta}_\lambda - \boldsymbol{\theta}^* \rangle, \quad (38)$$

which accounts explicitly for the cross term between stochastic estimation error and deterministic regularization bias. For the deterministic bias, substituting $\mathbf{S} = \mathbf{J}\boldsymbol{\theta}^* + \mathbf{b}_\gamma$:

$$\begin{aligned} \boldsymbol{\theta}_\lambda - \boldsymbol{\theta}^* &= (\mathbf{J} + \lambda \mathbf{I})^{-1} (\mathbf{S} - (\mathbf{J} + \lambda \mathbf{I})\boldsymbol{\theta}^*) \\ &= (\mathbf{J} + \lambda \mathbf{I})^{-1} (\mathbf{S} - \mathbf{J}\boldsymbol{\theta}^* - \lambda \boldsymbol{\theta}^*) \\ &= (\mathbf{J} + \lambda \mathbf{I})^{-1} (\mathbf{b}_\gamma - \lambda \boldsymbol{\theta}^*), \end{aligned} \quad (39)$$

which establishes the exact deterministic identity (39). Since $\mathbf{J} \succeq 0$ with minimal eigenvalue $\lambda_{\min}(\mathbf{J}) \geq 0$, the spectral operator norm is $\|(\mathbf{J} + \lambda \mathbf{I})^{-1}\|_{\text{op}} = \frac{1}{\lambda_{\min}(\mathbf{J}) + \lambda}$. Applying the triangle inequality:

$$\|\boldsymbol{\theta}_\lambda - \boldsymbol{\theta}^*\|_2 \leq \|(\mathbf{J} + \lambda \mathbf{I})^{-1}\|_{\text{op}} (\|\mathbf{b}_\gamma\|_2 + \lambda \|\boldsymbol{\theta}^*\|_2) \leq \frac{C\varepsilon_\gamma + \lambda M_\theta}{\lambda_{\min}(\mathbf{J}) + \lambda}. \quad (40)$$

For the stochastic variance, expanding $\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}_\lambda$:

$$\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}_\lambda = (\mathbf{J} + \lambda \mathbf{I})^{-1} \left(\hat{\mathbf{S}} - \mathbf{S} - (\hat{\mathbf{J}} - \mathbf{J})\boldsymbol{\theta}_\lambda \right) + o_p(N^{-1/2}). \quad (41)$$

By Theorem 5(a), $\sqrt{N}(\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}_\lambda) \Rightarrow \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_\lambda^{\text{tot}})$. Under uniform integrability of $N\|\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}_\lambda\|_2^2$, stochastic variance satisfies $V_N \equiv \mathbb{E}\|\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}_\lambda\|_2^2 = \frac{1}{N} \text{Tr}(\boldsymbol{\Sigma}_\lambda^{\text{tot}}) + o(N^{-1}) \leq \frac{\text{Tr}(\boldsymbol{\Sigma}_\lambda^{\text{tot}})}{N(\lambda_{\min}(\mathbf{J}) + \lambda)^2} + o(N^{-1})$, using $\text{Tr}(\mathbf{A}^{-1}\boldsymbol{\Omega}\mathbf{A}^{-1}) \leq \|\mathbf{A}^{-2}\|_{\text{op}} \text{Tr}(\boldsymbol{\Omega})$ for $\mathbf{A} = \mathbf{J} + \lambda \mathbf{I} \succ 0$. Applying Minkowski's inequality in $L_2(P; \mathbb{R}^K)$ to the decomposition $(\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}^*) = (\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}_\lambda) + (\boldsymbol{\theta}_\lambda - \boldsymbol{\theta}^*)$:

$$\left(\mathbb{E}\|\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}^*\|_2^2 \right)^{1/2} \leq \|\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}_\lambda\|_{L_2} + \|\boldsymbol{\theta}_\lambda - \boldsymbol{\theta}^*\|_2 \leq \sqrt{V_N} + B_\lambda, \quad (42)$$

which establishes the finite-sample risk upper bound:

$$\mathbb{E}\|\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}^*\|_2^2 \leq (B_\lambda + \sqrt{V_N})^2 \leq 2B_\lambda^2 + 2V_N \leq 2 \left(\frac{C\varepsilon_\gamma + \lambda M_\theta}{\lambda_{\min}(\mathbf{J}) + \lambda} \right)^2 + \frac{2\text{Tr}(\boldsymbol{\Sigma}_\lambda^{\text{tot}})}{N(\lambda_{\min}(\mathbf{J}) + \lambda)^2} + o(N^{-1}), \quad (43)$$

completely avoiding any requirement to bound the stochastic bias cross-term.

Risk Bound Minimizer vs. Empirical Grid Optimum: Differentiating the leading-order risk upper bound $R_{\text{bound}}(\lambda) \equiv \left(\frac{C\varepsilon_\gamma + \lambda M_\theta}{\lambda_{\min}(\mathbf{J}) + \lambda} \right)^2 + \frac{\text{Tr}(\boldsymbol{\Sigma}_\lambda^{\text{tot}})}{N(\lambda_{\min}(\mathbf{J}) + \lambda)^2}$ with respect to $\lambda \geq 0$, the bound-optimal shrinkage parameter $\lambda_{\text{bound}}^* \equiv \arg \min_{\lambda \geq 0} R_{\text{bound}}(\lambda)$ satisfies:

$$\frac{\partial}{\partial \lambda} \left[\left(\frac{C\varepsilon_\gamma + \lambda M_\theta}{\lambda_{\min}(\mathbf{J}) + \lambda} \right)^2 + \frac{\text{Tr}(\boldsymbol{\Sigma}_\lambda^{\text{tot}})}{N(\lambda_{\min}(\mathbf{J}) + \lambda)^2} \right] = 0. \quad (44)$$

When the overlap Gram matrix is ill-conditioned ($\lambda_{\min}(\mathbf{J}) \approx 0$), the variance term $\frac{\text{Tr}(\boldsymbol{\Sigma}_\lambda^{\text{tot}})}{N(\lambda_{\min}(\mathbf{J}) + \lambda)^2}$ dominates for small λ , while the deterministic shrinkage bias $\left(\frac{\lambda M_\theta}{\lambda_{\min}(\mathbf{J}) + \lambda} \right)^2$ dominates for large λ . The interior minimizer $\lambda_{\text{bound}}^* > 0$ balances variance reduction against shrinkage bias in the upper bound. We distinguish λ_{bound}^* and the empirical surrogate minimizer $\hat{\lambda}_{\text{emp}}^* = \arg \min_\lambda \hat{R}(\lambda)$ from the true infeasible risk minimizer $\lambda_{\text{true}}^* \equiv \arg \min_\lambda \mathbb{E}\|\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}^*\|_2^2$. In the ill-conditioned benchmark of Table 11 (100 replications, mean $\lambda_{\min} = 0.026$), the grid minimum of the empirical MSE is at $\lambda = 0.1$, a $55\times$ reduction relative to $\lambda = 0$, with coverage of the regularized target 87% (versus 1% unregularized). These are properties of that DGP, not constants. In the well-conditioned frontier (Table 2) the penalty is immaterial.

Table 11: Regularization path in an ill-conditioned design (`reports/or_dml_regularization_frontier.csv`).

λ	bias	variance	MSE	coverage (%)
0	0.8504	0.0687	0.7854	1
0.0001	0.8465	0.0684	0.7785	1
0.001	0.8131	0.0663	0.7210	1
0.005	0.6907	0.0583	0.5293	2
0.01	0.5784	0.0506	0.3795	4
0.02	0.4287	0.0395	0.2177	13
0.05	0.2122	0.0233	0.0615	46
0.1	0.0903	0.0143	0.0143	87
0.2	0.1259	0.0090	0.0209	56
0.5	0.3237	0.0048	0.1098	4

I.3 Derivation of Adaptive Spectral Regularization Parameter and Oracle Bound

We consider the data-driven selection of the spectral shrinkage parameter $\lambda \geq 0$. While classical Tikhonov regularization selects shrinkage via prediction cross-validation, in OR-DML the target is the structural causal parameter vector $\boldsymbol{\theta}^*$, where prediction cross-validation conflates causal identification with prognostic predictive fit. We distinguish two distinct regularization regimes:

1. **Inferential Regularization** ($\lambda = o(N^{-1/2})$): Required by Theorem 5 to guarantee asymptotic unbiasedness $\sqrt{N}\text{Bias}(\lambda) \rightarrow \mathbf{0}$, ensuring centered Gaussian limit laws and valid nominal confidence interval coverage.
2. **Finite-Sample Risk Minimization** ($\lambda_{\text{risk}} = \mathcal{O}(1)$): Minimizes the finite-sample expected ℓ_2 estimation risk $\mathbb{E}\|\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}^*\|_2^2$ under severe causal ill-conditioning ($\lambda_{\min}(\mathbf{J}) \approx 0$).

Empirical Risk Surrogate and Convexity: Define the empirical plug-in risk surrogate $\hat{R}(\lambda)$ by substituting empirical estimators $(\hat{\varepsilon}_\gamma, \lambda_{\min}(\hat{\mathbf{J}}), \hat{\boldsymbol{\Omega}}, \|\hat{\boldsymbol{\theta}}_0\|_2)$ into the risk upper bound of Theorem 6:

$$\hat{R}(\lambda) = \frac{(C\hat{\varepsilon}_\gamma + \lambda\|\hat{\boldsymbol{\theta}}_0\|_2)^2 + N^{-1}\text{Tr}(\hat{\boldsymbol{\Omega}})}{(\lambda_{\min}(\hat{\mathbf{J}}) + \lambda)^2}. \quad (45)$$

Let $u = \lambda_{\min}(\hat{\mathbf{J}}) + \lambda \geq \lambda_{\min}(\hat{\mathbf{J}}) > 0$. Rewriting $\lambda = u - \lambda_{\min}$, the numerator becomes:

$$(C\hat{\varepsilon}_\gamma - \lambda_{\min}(\hat{\mathbf{J}})\|\hat{\boldsymbol{\theta}}_0\|_2 + u\|\hat{\boldsymbol{\theta}}_0\|_2)^2 + N^{-1}\text{Tr}(\hat{\boldsymbol{\Omega}}) = (A + uB)^2 + V, \quad (46)$$

where $A = C\hat{\varepsilon}_\gamma - \lambda_{\min}(\hat{\mathbf{J}})\|\hat{\boldsymbol{\theta}}_0\|_2$, $B = \|\hat{\boldsymbol{\theta}}_0\|_2$, and $V = N^{-1}\text{Tr}(\hat{\boldsymbol{\Omega}})$. Dividing by u^2 :

$$\hat{R}(u - \lambda_{\min}) = B^2 + \frac{2AB}{u} + \frac{A^2 + V}{u^2}. \quad (47)$$

Differentiating with respect to u :

$$\frac{d\hat{R}}{du} = -\frac{2AB}{u^2} - \frac{2(A^2 + V)}{u^3} = -\frac{2}{u^3} [ABu + (A^2 + V)]. \quad (48)$$

Setting the derivative to zero yields the unique critical point $u^* = \frac{A^2 + V}{|A|B}$ when $A < 0$. When $A \geq 0$ (i.e., $C\hat{\varepsilon}_\gamma \geq \lambda_{\min}(\hat{\mathbf{J}})\|\hat{\boldsymbol{\theta}}_0\|_2$), $\frac{d\hat{R}}{du} < 0$ on $(0, \infty)$, so that the objective is strictly decreasing toward B^2 as $\lambda \rightarrow \infty$. Constraining the search to a compact admissible set $\lambda \in [0, \bar{\lambda}]$ with a fixed upper bound $\bar{\lambda} < \infty$ (independent of N), the bounded empirical minimizer is:

$$\hat{\lambda}^* = \begin{cases} \text{clip} \left(\frac{(C\hat{\varepsilon}_\gamma - \lambda_{\min}(\hat{\mathbf{J}})\|\hat{\boldsymbol{\theta}}_0\|_2)^2 + N^{-1}\text{Tr}(\hat{\boldsymbol{\Omega}})}{(\lambda_{\min}(\hat{\mathbf{J}})\|\hat{\boldsymbol{\theta}}_0\|_2 - C\hat{\varepsilon}_\gamma)\|\hat{\boldsymbol{\theta}}_0\|_2} - \lambda_{\min}(\hat{\mathbf{J}}), 0, \bar{\lambda} \right), & \text{if } A < 0, \\ \bar{\lambda}, & \text{if } A \geq 0. \end{cases} \quad (49)$$

Because $R'(u) = -\frac{2}{u^3} [ABu + (A^2 + V)]$ changes sign strictly from negative to positive at u^* when $A < 0$, u^* is the unique global interior minimizer on $(0, \infty)$.

Lemma 4 (Oracle Inequality for the Surrogate). *Let R_{surr} be the population surrogate, $\hat{R}$ its plug-in, and $\Delta_N = \sup_{\lambda \in [0, \bar{\lambda}]} |\hat{R}(\lambda) - R_{\text{surr}}(\lambda)| = \mathcal{O}_P(N^{-1/2})$. Then $R_{\text{surr}}(\hat{\lambda}^*) \leq \inf_{\lambda \in [0, \bar{\lambda}]} R_{\text{surr}}(\lambda) + 2\Delta_N$.*

Proof. For any $\lambda^\circ \in [0, \bar{\lambda}]$: $R_{\text{surr}}(\hat{\lambda}^*) \leq \hat{R}(\hat{\lambda}^*) + \Delta_N \leq \hat{R}(\lambda^\circ) + \Delta_N \leq R_{\text{surr}}(\lambda^\circ) + 2\Delta_N$. $\square$

This guarantee governs the surrogate, not the true risk; calibrated plug-in $\hat{\varepsilon}_\gamma = \frac{2}{N} \sum_t \text{Gini}(\gamma_t)$ preserves ranking when $C\hat{\varepsilon}_\gamma$ dominates. $\blacksquare$

J ALGORITHM AND SCALABILITY

Algorithm 1 (main text) lists the steps. While our primary experiments study binary regimes ($K = 2$), the OR-DML coupled estimator generalizes directly to arbitrary finite state dimensions $K \geq 2$. Here we formalize the computational and statistical scalability of the coupled Gram matrix $\hat{\mathbf{J}} \in \mathbb{R}^{K \times K}$.

Computational Complexity and Matrix Concentration under α -Mixing: Constructing $\hat{\mathbf{J}}$ and score $\hat{\mathbf{S}}$ requires evaluating K nuisance pairs across B folds ($\mathcal{O}(BK \cdot \text{Cost}(\text{Nuisance}))$). Forming $\hat{\mathbf{J}}$ takes $\mathcal{O}(NK^2)$ operations, and solving $(\hat{\mathbf{J}} + \lambda \mathbf{I})^{-1} \hat{\mathbf{S}}$ via Cholesky factorization takes $\mathcal{O}(K^3)$ operations, taking < 1 ms for

$K \leq 20$. Statistically, the empirical Gram matrix $\hat{\mathbf{J}} = \frac{1}{N} \sum_{t=1}^N \mathbf{\Gamma}_t \tilde{\mathbf{T}}_t \tilde{\mathbf{T}}_t^\top \mathbf{\Gamma}_t$ concentrates around $\mathbf{J}$ under geometric α -mixing with spectral gap $\rho_{\text{gap}} \equiv 1 - |\lambda_2(\mathbf{\Pi})| > 0$ via the Matrix Bernstein inequality (Banna et al., 2016):

$$\|\hat{\mathbf{J}} - \mathbf{J}\|_{\text{op}} \leq C_J \sqrt{\frac{K \log(K/\delta)}{N \cdot \rho_{\text{gap}}}} + \frac{C'_J K \log(K/\delta)}{N \cdot \rho_{\text{gap}}}. \quad (50)$$

By Weyl’s theorem, $|\lambda_{\min}(\hat{\mathbf{J}}) - \lambda_{\min}(\mathbf{J})| \leq \|\hat{\mathbf{J}} - \mathbf{J}\|_{\text{op}}$, ensuring that $\lambda_{\min}(\hat{\mathbf{J}})$ concentrates at the standard $\mathcal{O}_P(N^{-1/2})$ rate whenever $\lambda_{\min}(\mathbf{J}) > 0$.

Non-Identifiability vs. Shrinkage: When Systems Must Abstain: Spectral shrinkage $\hat{\theta}_\lambda = (\hat{\mathbf{J}} + \lambda \mathbf{I})^{-1} \hat{\mathbf{S}}$ stabilizes variance under ill-conditioned overlap ($\lambda_{\min} \approx 0$). However, when data lack identifying variation entirely—diagnosed by condition number $\kappa(\hat{\mathbf{J}}) > 10^4$ or a moving-block bootstrap 95% confidence interval for $\lambda_{\min}(\hat{\mathbf{J}})$ that covers zero—the structural parameter θ^* suffers catastrophic ill-conditioning (and structural non-identification in the exact limit $\lambda_{\min} = 0$, Theorem 2). In this regime, heavy shrinkage $\lambda \rightarrow \infty$ artificially forces $\|\hat{\theta}_\lambda\| \rightarrow 0$ without recovering structural effects. Deployed decision pipelines should treat $\kappa(\hat{\mathbf{J}}) > 10^4$ as an operational abstention trigger rather than masking severe ill-conditioning with excessive shrinkage. Table 12 reports OR-DML with oracle-quality proxies for $K \in \{2, 3, 4, 5\}$ ($N = 1200$).

Table 12: Multi-regime scalability (`reports/multiregime_scalability_summary.csv`).

K	ATE bias	ATE SE	$\ \hat{\theta} - \theta\ _2$	$\lambda_{\min}$	κ
2	0.062	0.069	0.112	0.758	1.31
3	0.074	0.062	0.142	0.506	1.22
4	0.036	0.073	0.193	0.361	1.43
5	0.061	0.073	0.389	0.238	2.00

K FACTORIAL AND LEARNED-POSTERIOR EXPERIMENTS

Factorial design (`src/factorial_reliability_experiment.py`). A two-state chain with persistence 0.88 and exit probability 0.15 generates $T = 2(1-S) + 5S + 0.8X_0 - 0.5X_1 + \sigma_S V$ with $\sigma_0 = 1$ and $\sigma_1 \in \{0.08, \dots, 1.5\}$, and $Y = \theta_S T + 15(1-S) + 50S + 1.2X_0 + 0.9X_2 + U$ with $\theta = (0.75, 2.5)$ and $N = 1200$. Nuisances are cross-fitted with oracle regime weights; the proxy is the label mixture $\gamma = (1 - \varepsilon_\gamma/2)\mathbf{H} + (\varepsilon_\gamma/2)(\mathbf{1} - \mathbf{H})$ for $\varepsilon_\gamma \in \{0, \dots, 0.4\}$, with 100 sequences per cell (4,900 runs). Rank correlations: run-level 0.995 for $\varepsilon_\gamma/\lambda_{\min}$, 0.626 for ε_γ , 0.521 for $1/\lambda_{\min}$, 0.535 for the additive score; cell-level 0.996. Retaining the 40% of runs with lowest $\varepsilon_\gamma/\lambda_{\min}$ reduces mean error from 48.81 to 2.61. Because the proxy is a deterministic corruption of the true state, these numbers verify Proposition 1 rather than a statistical property of learned representations. Rerunning the script reproduces the committed raw data to within 10^{-12} .

Numerator scale. In a feature-branch exploration with proxy error fixed at 0.008 and $\lambda_{\min}$ essentially flat (0.44–0.50), increasing the treatment shift from 0.5 to 8 raises the causal error from 0.18 to 1.89. The ratio $\varepsilon_\gamma/\lambda_{\min}$ omits the regime-shift scale that multiplies state error in both laws.

Learned posteriors (`src/synthesis/learned_posterior_coupled.py`). Same structural equations, with treatment shift $\Delta_T \in \{1.5, 3, 6\}$, $\sigma_1 \in \{0.3, 0.75, 1.5\}$ and 2-d Gaussian proxies with Mahalanobis separation $\Delta_Z \in \{0.75, 1.25, 2, 3\}$; full-covariance HMM (three restarts, smoothing); posterior-weighted Ridge nuisances on five purged folds (embargo 12); 20 replications per cell. The oracle Gram uses the true one-hot weights with the same nuisance procedure. The plug-in shifts $\hat{\Delta}_T, \hat{\Delta}_Y$ are posterior-weighted differences of regime means of T and Y . Raw output: `reports/synthesis/learned_posterior_coupled_raw.csv`.

L SYNTHETIC FRONTIERS

Observability and persistence frontier (`src/synthesis/frontier_designs.py`). The DGP is the difficulty-frontier generator (`src/synthetic_dgp_benchmark.py`): two regimes, $\theta = (0.75, 2.5)$, treatment shift 4, outcome offsets 15/50, AR(1) controls, 2-d Gaussian proxies with separation Δ_Z , $N = 1200$, stationary ATE 1.53. Posteriors are learned by a full-covariance Gaussian HMM (EM, three restarts, smoothing) from Z alone. All

eight estimators share the posteriors, five purged blocks (embargo 12), Ridge nuisances and a HAC lag of 12. Labels are aligned to the truth only to evaluate regime-specific errors. The grid is $\Delta_Z \in \{0.2, 0.5, 1, 2, 4\}$ at regime-0 stay probability 0.88, plus stay probabilities $\{0.70, 0.97\}$ at $\Delta_Z \in \{1, 2\}$, with 100 datasets per cell (900 in all). The DGP deliberately violates proxy sufficiency (the treatment carries state information), and $\Delta_Z \leq 0.5$ is close to the non-identification boundary of Theorem 2, where every proxy-based estimator fails.

Legacy difficulty benchmark (`src/synthetic_dgp_benchmark.py`). The earlier eleven-method benchmark on the same DGP (Table 15) includes forward filtering, decoupled and unregularized variants, GBM proxy controls and HMM(Z, T); its “Spectral OR-DML” rows are the posterior-weighted design. HMM(Z, T), which infers the state from (Z, T) , is best at low separation because the treatment carries state information. The $\lambda_{\min}$ column displays phantom resolution directly: residualizing on X only gives 4.95, against 0.45 for the oracle state, and at $\Delta_Z = 0.2$ the weighted design’s generated-state value is 0.80 with $\varepsilon_\gamma = 0.78$. **Frontier B (size)**. With $\theta_1^* = 0$, nominal 5% Wald tests reject in 78.0% of runs for the weighted design and 31.0% for hard fixed effects: the sandwich is centered at the proxy target θ_γ , not at θ^* (Theorem 5). **Soft versus hard under a continuous latent mixture** (`src/test_soft_vs_hard_continuous_mixture.py`). With a continuous latent mixture weight and an informative proxy, the posterior-weighted coupled estimator has mean absolute bias 3.71, hard fixed effects 2.81 and standard DML 3.83. Soft weighting wins in 0% of 100 replications; the experiment was written to exhibit an advantage of soft weighting and found none for that design. With posterior-adjusted nuisances on the same inputs, OR-DML has bias 2.32 and beats hard fixed effects in 100% of replications: the posterior helps once it is used as an adjustment feature rather than a nuisance weight.

M REAL-DATA-CALIBRATED SEMI-SYNTHETIC BENCHMARK

`src/synthesis/semisynthetic_hourly.py`. For each city, the analysis sample consists of the hours on the corrected grid where NO_2 , $\text{PM}_{2.5}$, the proxies and the controls are all observed. The real missingness pattern is therefore inherited, including outages and the shorter Kolkata record. The proxies are temperature, humidity and wind; the controls are pressure, diurnal phase and lagged NO_2 . A two-state diagonal Gaussian HMM (five restarts) is calibrated on the standardized proxies. In each of 30 replications the true path is drawn from that HMM’s exact smoothing distribution by forward-filtering backward-sampling, so regime durations and overlap are the calibrated ones. Then $T = \text{NO}_2^{\text{std}} + S$ and $Y = \theta_S T + 3S + X\beta + U$, with $\theta = (0.5, 1.5)$ and U an AR(1) process with coefficient 0.8. Estimators receive the calibrated posteriors, the most favourable case for posterior methods. Besides the estimators of Table 2, the “Spectral OR-DML (soft, coupled)” rows are the posterior-weighted design and “OR-DML (posterior-adjusted)” is the paper’s estimator. The bias-law column uses the heterogeneous-effects form with the observed $\hat{v}$ and $\widehat{\text{Var}}(\tilde{T})$ and the injected (Δ_T, Δ_g) . Bias-law prediction versus observed mean bias of posterior-feature DML: Delhi 0.36 vs 0.55; Mumbai 1.41 vs 1.03; Bengaluru 1.70 vs 1.51; Kolkata 2.00 vs 2.17. The residual gaps reflect restricted (linear) learners and the assumption $\mathbb{E}[m_0(X)\text{Var}(D | F)] = 0$, which the real NO_2 series need not satisfy. The oracle-state intervals under-cover in some cities (70–93%), which we attribute to the HAC truncation at 24 hours under strongly persistent real NO_2 .

N REPRESENTATIONS, CALIBRATION AND RELIABILITY

LatentRegimeBench (`src/train_real_representation_zoo.py`, `src/calibration_intervention_zoo.py`, `src/hierarchical_reliability_regression.py`) trains a Gaussian HMM (causal filter), a GRU, a causal Transformer and a linear SSM on shifted worlds ($N = 2000$, five seeds), and evaluates state recovery, calibration and downstream failure discrimination of the rolling score $\mathcal{D}_t = H(\gamma_t)/\lambda_{\min}(\mathbf{J}_t)$. The HMM attains failure ROC-AUC 0.615. That is no better than plain entropy, and the neural encoders are near chance. Consistent with Theorem 4, the representation’s *probabilistic* quality (NLL) predicts reliability across worlds, but recalibration, which changes $\hat{v}$ and not the conditional information, does not improve it.

O SENSOR DATA, AUDIT AND FULL CITY RESULTS

Sources. Hourly pollutant concentrations come from CPCB continuous monitoring stations via OpenAQ (OpenAQ, 2025; CPCB, 2025), and weather from Open-Meteo reanalysis (Open-Meteo, 2025). The record runs from February 2025 to June 2026 (Kolkata to June 2025), and raw files are in `data/raw_hourly`. **Time-grid audit.** OpenAQ stamps readings at `hh:30Z`. The legacy pipeline (`src/data_pipeline_clean.py`) rounded these

with ties-to-even, so every reading mapped to an even hour, and the inner join with hourly weather kept only even hours. The legacy “hourly” panel is in fact two-hourly: lag-1 is two hours and its rolling 3-hour mean covers six. The corrected grid (`src/bias_low/ext_real_hourly_fix.py`) floors readings to the hour, pairing each reading with the weather hour that began 30 minutes earlier (a causal alignment). It reindexes to a regular hourly grid, forward-fills at most two hours, and rebuilds lags on the grid. At least 93% of consecutive analysis rows are exactly one hour apart in every city. All real-data results in the main text use the corrected grid. Legacy two-hourly results are retained in `reports/` for provenance only. **Aligned placebos.** For a lead $h < 0$ we regress $\text{PM}_{2.5,t+h}$ on $\text{NO}_{2,t}$ with the controls and contemporaneous NO_2 measured at the outcome time $t + h$, so the test asks whether future NO_2 predicts past $\text{PM}_{2.5}$ beyond the information available then. Without this alignment, the $h = -1$ placebo is identically zero because $\text{PM}_{2.5,t-1}$ is a control.

P REPRODUCTION AND VERIFICATION

One command. `bash reproduce.sh` reruns, in order:

1. the synthesis experiments, about ten minutes on two CPU cores: the closed-form verification, the coupled-law and orthogonality checks, the observability–persistence frontier, learned posteriors, the semi-synthetic benchmark, the city stress test, the HAC lag sensitivity and soft-versus-hard;
2. the factorial experiment (about 20 s);
3. the figures, the generated tables and `numbers.tex`;
4. `verify_submission.py`.

`bash reproduce.sh --full` also reruns the longer legacy benchmarks: the difficulty frontier, the bias-law suite and the representation zoo. **Single source of numbers.** Every numeric claim in the main text is a $\text{\LaTeX}$ macro defined in `paper/generated/numbers.tex`, and every results table is an `\input` of `paper/generated/tab_*.tex`. Both are written by `src/synthesis/make_tables_numbers.py` from committed CSV/JSON outputs. `paper/generated/claims.csv` lists each macro with its value, source file and derivation. **Verifier.** `verify_submission.py` (i) regenerates all macros and tables into a temporary directory and requires byte-identical output, (ii) re-evaluates the closed form and its symbolic identities, (iii) checks that every macro used in the manuscript is defined and every defined macro is used or listed, (iv) checks that the PDF is newer than its sources and has an eight-page main body, and (v) scans the PDF and the supplementary archive for de-anonymizing strings.

Q LINEAGE, CORRECTIONS AND SCOPE

This paper consolidates several development lines (an applied study, a regime-conditional DML draft, an overlap-aware DML draft, a research ledger on task-relative reliability, and a residual-state bias-law analysis). Audits across them changed the claims as follows. (1) The two-hour time grid (Appendix O) invalidated “hourly” labels, horizons and lag lengths in earlier drafts; all real-data numbers here are recomputed. (2) Earlier placebo leads were not aligned to the outcome’s information set; the one-hour lead was mechanically zero. (3) Earlier drafts presented $\varepsilon_\gamma/\lambda_{\min}$ as a general reliability predictor on the basis of the factorial design; Proposition 1 shows that design’s correlation is algebraic, and learned posteriors reduce it to 0.517. (4) Earlier drafts used posterior-*weighted* regime nuisances in the coupled estimator and presented it as the method of choice; every benchmark here favours hard fixed effects over that design. Proposition 1 locates the failure in the nuisance step, and the posterior-adjusted design of Lemma 1 replaces it. (5) A semi-synthetic benchmark built on the legacy grid, in which the true state was a deterministic function of the proxies, is superseded by Appendix M. (6) A synthetic financial transfer experiment, an imputation-sensitivity table and a multi-horizon failure-forecasting table built on the legacy grid, and an early policy simulation with an unsupported mortality scaling, are not part of this paper’s evidence; their code remains in the archive for transparency. (7) Regularization numbers are recomputed from the committed path (Table 11). The full research ledger, earlier manuscripts and changelogs are in the supplementary archive under `archive/` and `research/`.

Table 13: Observability $\times$ persistence frontier, all estimators (reports/synthesis/frontier_designs_summary.csv). Coverage: nominal 95% intervals for the sample ATE. $\|\hat{\theta} - \theta^*\|$: median over datasets. $\lambda_{\min}$: generated-state Gram matrix (median).

Δ_Z	p_{00}	Estimator	mean bias	median	RMSE	cover.	$\ \hat{\theta} - \theta^*\ $	$\lambda_{\min}$
0.2	0.88	Oracle-state	0.029	0.027	0.035	95%	0.064	0.435
		Standard DML	8.442	8.445	8.442	0%	—	—
		Block DML	8.472	8.470	8.473	0%	—	—
		DML + Z controls	8.291	8.292	8.292	0%	—	—
		Posterior-feature DML	8.162	8.156	8.165	0%	—	—
		Hard regime FE	8.240	8.231	8.242	0%	11.347	1.076
		OR-DML, weighted nuisances	8.634	8.646	8.636	0%	12.144	0.460
		OR-DML (posterior-adjusted)	8.126	8.115	8.129	0%	11.230	0.404
0.5	0.88	Oracle-state	0.028	0.024	0.034	95%	0.058	0.451
		Standard DML	8.421	8.418	8.422	0%	—	—
		Block DML	8.455	8.456	8.455	0%	—	—
		DML + Z controls	7.427	7.408	7.428	0%	—	—
		Posterior-feature DML	6.009	5.996	6.014	0%	—	—
		Hard regime FE	6.460	6.461	6.465	0%	9.109	1.119
		OR-DML, weighted nuisances	9.025	9.068	9.029	0%	12.865	0.760
		OR-DML (posterior-adjusted)	5.940	5.937	5.946	0%	8.446	0.554
1.0	0.88	Oracle-state	0.026	0.025	0.033	96%	0.052	0.452
		Standard DML	8.432	8.444	8.433	0%	—	—
		Block DML	8.465	8.463	8.466	0%	—	—
		DML + Z controls	5.433	5.440	5.435	0%	—	—
		Posterior-feature DML	2.164	2.080	2.280	0%	—	—
		Hard regime FE	2.575	2.489	2.674	0%	3.652	0.597
		OR-DML, weighted nuisances	4.236	4.210	4.282	0%	6.107	0.547
		OR-DML (posterior-adjusted)	2.134	2.046	2.252	0%	2.954	0.498
2.0	0.88	Oracle-state	0.031	0.027	0.038	94%	0.060	0.448
		Standard DML	8.437	8.437	8.438	0%	—	—
		Block DML	8.474	8.476	8.475	0%	—	—
		DML + Z controls	2.633	2.640	2.637	0%	—	—
		Posterior-feature DML	0.141	0.045	0.865	98%	—	—
		Hard regime FE	0.141	0.042	0.852	93%	0.077	0.453
		OR-DML, weighted nuisances	0.147	0.045	0.867	94%	0.095	0.451
		OR-DML (posterior-adjusted)	0.129	0.038	0.860	91%	0.070	0.449
4.0	0.88	Oracle-state	0.031	0.027	0.037	94%	0.063	0.445
		Standard DML	8.444	8.445	8.444	0%	—	—
		Block DML	8.481	8.484	8.481	0%	—	—
		DML + Z controls	0.864	0.862	0.870	0%	—	—
		Posterior-feature DML	0.044	0.036	0.056	100%	—	—
		Hard regime FE	0.031	0.027	0.037	94%	0.063	0.445
		OR-DML, weighted nuisances	0.032	0.028	0.037	94%	0.063	0.445
		OR-DML (posterior-adjusted)	0.031	0.028	0.037	94%	0.063	0.445
1.0	0.70	Oracle-state	0.027	0.024	0.033	95%	0.064	0.337
		Standard DML	7.961	7.945	7.962	0%	—	—
		Block DML	7.978	7.954	7.979	0%	—	—
		DML + Z controls	5.174	5.183	5.177	0%	—	—
		Posterior-feature DML	2.742	2.726	2.770	0%	—	—
		Hard regime FE	3.260	3.230	3.290	0%	5.220	0.546
		OR-DML, weighted nuisances	5.256	5.260	5.280	0%	8.811	0.470
		OR-DML (posterior-adjusted)	2.725	2.705	2.751	0%	4.467	0.409
2.0	0.70	Oracle-state	0.027	0.023	0.034	96%	0.060	0.335
		Standard DML	7.959	7.953	7.960	0%	—	—
		Block DML	7.979	7.971	7.980	0%	—	—
		DML + Z controls	2.521	2.515	2.524	0%	—	—
		Posterior-feature DML	0.060	0.049	0.083	98%	—	—
		Hard regime FE	0.070	0.039	0.101	97%	0.084	0.336
		OR-DML, weighted nuisances	0.079	0.048	0.112	94%	0.123	0.335
		OR-DML (posterior-adjusted)	0.052	0.032	0.073	96%	0.076	0.336
1.0	0.97	Oracle-state	0.025	0.021	0.033	93%	0.078	0.175
		Standard DML	7.609	7.641	7.616	0%	—	—
		Block DML	7.679	7.677	7.686	0%	—	—
		DML + Z controls	5.228	5.244	5.231	0%	—	—
		Posterior-feature DML	1.060	1.072	1.114	1%	—	—
		Hard regime FE	1.194	1.182	1.258	3%	2.483	0.225
		OR-DML, weighted nuisances	1.933	1.956	1.976	0%	5.088	0.211
		OR-DML (posterior-adjusted)	0.960	0.933	1.010	0%	2.282	0.197
2.0	0.97	Oracle-state	0.027	0.024	0.034	93%	0.074	0.159
		Standard DML	7.486	7.530	7.496	0%	—	—
		Block DML	7.556	7.577	7.567	0%	—	—
		DML + Z controls	2.639	2.632	2.642	0%	—	—
		Posterior-feature DML	0.051	0.039	0.069	97%	—	—
		Hard regime FE	0.039	0.025	0.059	94%	0.087	0.159
		OR-DML, weighted nuisances	0.041	0.028	0.059	94%	0.098	0.159
		OR-DML (posterior-adjusted)	0.035	0.024	0.051	93%	0.086	0.159

Table 14: Persistence: mean absolute ATE bias by regime-0 stay probability p_{00} (reports/synthesis/frontier_designs_summary.csv).

p_{00}	Δ_Z	switches	ε_γ	Post.-feature	Hard FE	OR-DML wtd.	OR-DML	oracle
0.70	1	239	0.090	2.742	3.260	5.256	2.725	0.027
0.70	2	243	0.001	0.060	0.070	0.079	0.052	0.027
0.88	1	158	0.062	2.164	2.575	4.236	2.134	0.026
0.88	2	159	0.000	0.141	0.141	0.147	0.129	0.031
0.97	1	62	0.026	1.060	1.194	1.933	0.960	0.025
0.97	2	58	0.000	0.051	0.039	0.041	0.035	0.027

Table 15: Legacy difficulty benchmark, all methods (reports/or_dml_benchmark_summary.csv). Coverage is for the sample/population ATE.

Δ_Z	Estimator	mean bias	median	RMSE	cov. S/P (%)	$\lambda_{\min}$
0.2	Oracle DML	0.056	0.044	0.070	96/94	0.449
	Standard DML	8.456	8.456	8.456	0/0	4.934
	Block DML	8.490	8.479	8.491	0/0	5.011
	DML + Z Controls (Ridge)	8.307	8.312	8.308	0/0	4.627
	DML + Z Controls (GBM)	8.018	8.025	8.019	0/0	5.325
	Regime FE DML (Hard)	8.318	8.324	8.319	0/0	1.745
	Unregularized Coupled DML	8.637	8.615	8.638	0/0	0.804
	Spectral OR-DML	8.627	8.604	8.628	0/0	0.804
	Filtered OR-DML	8.604	8.582	8.605	0/0	0.800
	Decoupled Soft OR-DML	8.381	8.381	8.381	0/0	1.534
	HMM(Z, T) OR-DML	6.504	6.606	6.546	0/0	0.487
0.5	Oracle DML	0.055	0.044	0.071	92/95	0.448
	Standard DML	8.429	8.432	8.430	0/0	4.956
	Block DML	8.467	8.483	8.468	0/0	5.041
	DML + Z Controls (Ridge)	7.429	7.424	7.430	0/0	3.401
	DML + Z Controls (GBM)	7.031	7.020	7.034	0/0	3.819
	Regime FE DML (Hard)	6.628	6.672	6.634	0/0	1.203
	Unregularized Coupled DML	8.848	8.887	8.851	0/0	0.828
	Spectral OR-DML	8.835	8.874	8.839	0/0	0.828
	Filtered OR-DML	8.988	9.003	8.991	0/0	0.875
	Decoupled Soft OR-DML	7.164	7.178	7.167	0/0	1.311
	HMM(Z, T) OR-DML	4.324	4.390	4.364	0/0	0.482
1.0	Oracle DML	0.057	0.043	0.073	92/92	0.452
	Standard DML	8.433	8.449	8.434	0/0	4.954
	Block DML	8.466	8.475	8.467	0/0	5.027
	DML + Z Controls (Ridge)	5.438	5.462	5.440	0/0	2.078
	DML + Z Controls (GBM)	3.633	3.642	3.647	0/0	1.885
	Regime FE DML (Hard)	2.647	2.700	2.685	0/0	0.609
	Unregularized Coupled DML	4.025	4.030	4.045	0/0	0.557
	Spectral OR-DML	4.017	4.022	4.037	0/0	0.557
	Filtered OR-DML	5.172	5.185	5.185	0/0	0.605
	Decoupled Soft OR-DML	3.294	3.286	3.310	0/0	0.670
	HMM(Z, T) OR-DML	1.196	1.192	1.230	0/0	0.460
2.0	Oracle DML	0.052	0.042	0.066	93/97	0.454
	Standard DML	8.432	8.451	8.433	0/0	4.954
	Block DML	8.461	8.472	8.462	0/0	5.024
	DML + Z Controls (Ridge)	2.631	2.614	2.635	0/0	1.351
	DML + Z Controls (GBM)	0.207	0.194	0.243	72/72	1.214
	Regime FE DML (Hard)	0.077	0.063	0.102	94/99	0.455
	Unregularized Coupled DML	0.082	0.073	0.105	93/97	0.455
	Spectral OR-DML	0.081	0.072	0.104	92/97	0.455
	Filtered OR-DML	0.102	0.081	0.131	92/94	0.456
	Decoupled Soft OR-DML	0.079	0.069	0.101	94/98	0.456
	HMM(Z, T) OR-DML	0.059	0.045	0.074	93/96	0.454
4.0	Oracle DML	0.054	0.048	0.067	96/95	0.439
	Standard DML	8.451	8.461	8.452	0/0	4.912
	Block DML	8.486	8.495	8.487	0/0	4.996
	DML + Z Controls (Ridge)	0.848	0.842	0.853	0/0	1.095
	DML + Z Controls (GBM)	0.101	0.082	0.125	89/89	1.189
	Regime FE DML (Hard)	0.054	0.048	0.067	96/95	0.439
	Unregularized Coupled DML	0.054	0.048	0.067	96/95	0.439
	Spectral OR-DML	0.054	0.046	0.067	96/95	0.439
	Filtered OR-DML	0.054	0.046	0.067	96/95	0.439
	Decoupled Soft OR-DML	0.054	0.046	0.067	96/95	0.439
	HMM(Z, T) OR-DML	0.054	0.046	0.067	96/95	0.439

Table 16: Semi-synthetic benchmark, all estimators (`reports/synthesis/semisynthetic_hourly_summary.csv`). Coverage: nominal 95% intervals for the sample ATE.

City	Estimator	bias	RMSE	cover.	$\ \hat{\theta} - \theta\ $	$\lambda_{\min}$ prx/orc
Delhi	Oracle-state regime DML	0.028	0.034	83%	0.05	
	Standard DML (random folds)	1.942	1.942	0%	–	
	Block DML (purged folds)	2.163	2.163	0%	–	
	DML + Z controls	1.222	1.222	0%	–	
	Posterior-feature DML	0.551	0.552	0%	–	
	Hard regime FE DML	0.549	0.550	0%	0.77	
	Spectral OR-DML (soft, coupled)	0.879	0.880	0%	1.21	0.085/0.082
	OR-DML (posterior-adjusted)	0.417	0.420	0%	0.59	0.085/0.082
Mumbai	Oracle-state regime DML	0.073	0.094	83%	0.16	
	Standard DML (random folds)	3.004	3.005	0%	–	
	Block DML (purged folds)	3.088	3.089	0%	–	
	DML + Z controls	3.027	3.028	0%	–	
	Posterior-feature DML	1.028	1.032	0%	–	
	Hard regime FE DML	1.861	1.863	0%	2.84	
	Spectral OR-DML (soft, coupled)	3.102	3.103	0%	4.59	0.024/0.010
	OR-DML (posterior-adjusted)	1.651	1.653	0%	2.56	0.024/0.010
Bengaluru	Oracle-state regime DML	0.116	0.134	70%	0.22	
	Standard DML (random folds)	2.971	2.971	0%	–	
	Block DML (purged folds)	3.203	3.204	0%	–	
	DML + Z controls	2.602	2.603	0%	–	
	Posterior-feature DML	1.513	1.515	0%	–	
	Hard regime FE DML	1.973	1.975	0%	2.83	
	Spectral OR-DML (soft, coupled)	3.219	3.220	0%	4.56	0.019/0.011
	OR-DML (posterior-adjusted)	1.690	1.692	0%	2.40	0.019/0.011
Kolkata	Oracle-state regime DML	0.136	0.179	93%	0.33	
	Standard DML (random folds)	3.228	3.230	0%	–	
	Block DML (purged folds)	3.298	3.301	0%	–	
	DML + Z controls	3.055	3.058	0%	–	
	Posterior-feature DML	2.165	2.168	0%	–	
	Hard regime FE DML	2.300	2.304	0%	3.21	
	Spectral OR-DML (soft, coupled)	3.813	3.816	0%	5.33	0.018/0.008
	OR-DML (posterior-adjusted)	2.053	2.058	0%	2.88	0.018/0.008

Table 17: Temperature scaling (`reports/representation_zoo_calibration_intervention.csv`).

Encoder	T^*	ECE raw→cal.	NLL raw→cal.	AUC($\mathcal{D}_t$) raw→cal.
Gaussian HMM (Causal Filter)	1.00	0.0416→0.0416	0.205→0.205	0.5895→0.5895
Neural GRU Encoder	3.20	0.1050→0.0569	0.547→0.331	0.5644→0.5609
Causal Transformer Encoder	4.80	0.1379→0.0541	1.006→0.382	0.5356→0.5282
Linear State Space Model (SSM)	1.54	0.0496→0.0386	0.325→0.302	0.5492→0.5463

Table 18: Leave-one-world-out prediction of downstream reliability (15 worlds, `reports/representation_zoo_lowo_evaluation.csv`). With 15 worlds the effective sample is small; we report the comparison, not a significance claim.

LOWO predictors	RMSE	R^2	Spearman ρ	p
F_1	0.0525	0.113	0.374	3.2e-03
F_1 +ECE	0.0513	0.152	0.439	4.5e-04
F_1 +NLL	0.0492	0.221	0.526	1.6e-05
F_1 +ECE+NLL	0.0498	0.204	0.495	5.7e-05

Table 19: Dynamic responses and lead placebos by horizon (hours; ATE, *: $p < 0.05$; `reports/synthesis/city_placebo_hourly.csv`). Delhi’s response peaks at 1.61 at $h = 6$ h, but its one-hour lead rejects, so we do not interpret the curve causally.

City	screen	$h = -6$	$h = -3$	$h = -1$	$h = 0$	$h = 1$	$h = 3$	$h = 6$	$h = 12$	$h = 24$
Delhi	no	-0.02	0.02	0.07*	0.23*	0.70*	1.49*	1.61*	0.81*	0.33*
Mumbai	no	-0.35	-0.47	1.79	3.02	5.68*	2.73	-5.46*	-4.94	-9.90*
Bengaluru	no	-0.03*	-0.04*	-0.05	0.00	0.01	-0.01	0.01	-0.05	-0.10
Kolkata	no	-0.04	-0.17	0.91	1.22	1.05	-0.82	-2.71	-2.24	0.30
Delhi	yes	-0.02	0.02	0.07*	0.23*	0.70*	1.49*	1.61*	0.81*	0.33*
Mumbai	yes	-9.28	3.16	5.42	0.38	4.17	-3.47	-4.52	-7.70	-15.61*
Bengaluru	yes	-0.04*	-0.05*	-0.06	0.02	0.03	0.00	-0.01	-0.08	-0.13*
Kolkata	yes	0.39	0.92	3.16*	3.32*	4.14*	1.77	3.88	17.20*	36.04*

Table 20: Newey–West lag sensitivity of regime-specific OR-DML with posterior-adjusted nuisances (corrected grid, frozen-sensor screen; `reports/bias_law/e6_hac_lag_sensitivity.csv`).

City	regime	lag 12	lag 24	lag 72	lag 168
Delhi	1	0.430 (0.050; $p=0.000$)	0.430 (0.053; $p=0.000$)	0.430 (0.058; $p=0.000$)	0.430 (0.065; $p=0.000$)
	2	0.102 (0.033; $p=0.002$)	0.102 (0.033; $p=0.002$)	0.102 (0.036; $p=0.005$)	0.102 (0.040; $p=0.012$)
Mumbai	1	-4.012 (4.968; $p=0.419$)	-4.012 (4.688; $p=0.392$)	-4.012 (4.872; $p=0.410$)	-4.012 (5.227; $p=0.443$)
	2	2.757 (1.473; $p=0.061$)	2.757 (1.496; $p=0.065$)	2.757 (1.547; $p=0.075$)	2.757 (1.622; $p=0.089$)
Bengaluru	1	0.182 (0.149; $p=0.224$)	0.182 (0.155; $p=0.242$)	0.182 (0.184; $p=0.323$)	0.182 (0.194; $p=0.348$)
	2	-0.158 (0.066; $p=0.017$)	-0.158 (0.070; $p=0.024$)	-0.158 (0.081; $p=0.051$)	-0.158 (0.092; $p=0.086$)
Kolkata	1	-0.027 (0.525; $p=0.959$)	-0.027 (0.550; $p=0.961$)	-0.027 (0.629; $p=0.966$)	-0.027 (0.613; $p=0.965$)
	2	7.084 (3.450; $p=0.040$)	7.084 (3.400; $p=0.037$)	7.084 (3.282; $p=0.031$)	7.084 (3.267; $p=0.030$)

Table 21: Sensitivity analysis on the corrected hourly grid (posterior-feature estimate, 90% block-bootstrap intervals; `reports/bias_law/x23_real_hourly_fix.csv`).

City	screened	$\hat{\theta}$ [90% CI]	b^* (SD) [90% CI]	$b^*/ \hat{c}_Y $
Delhi	no	0.37 [0.29, 0.44]	82.3 [54.3, 144]	560
Mumbai	no	2.23 [-0.66, 4.89]	11.9 [1.3, 43]	392
Bengaluru	no	0.01 [-0.11, 0.14]	26.8 [3.0, 445]	158
Kolkata	no	1.79 [-0.74, 4.60]	10.9 [1.3, 105]	35
Mumbai	yes	0.31 [-2.80, 2.88]	1.2 [0.5, 18]	39
Bengaluru	yes	0.02 [-0.12, 0.16]	35.3 [3.3, 375]	189
Kolkata	yes	4.53 [1.48, 7.42]	16.5 [4.2, 60]	38